

Commitment Schemes Based on Module-LIP

Hengyi Luo^{1,2}, Kaijie Jiang³, Renjie Jin⁵, Yanbin Pan^{1,2(✉)}, and Anyu Wang^{3,4}

¹ State Key Laboratory of Mathematical Sciences, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, China
luohengyi23@mails.ucas.ac.cn, panyanbin@amss.ac.cn

² School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing, China

³ Institute for Advanced Study, BNRist, Tsinghua University, Beijing, China
jkj21@mails.tsinghua.edu.cn, anyuwang@tsinghua.edu.cn

⁴ Zhongguancun Laboratory, Beijing, China

⁵ College of Science, National University of Defense Technology, Changsha, China
jinrj97@163.com

Abstract. At EUROCRYPT 2025, Jiang et al. proposed a universal framework for constructing commitment schemes via group actions, and instantiated them with the Lattice Isomorphism Problem (LIP) and the Lattice Automorphism Problem (LAP), which marked the first application of LIP and LAP in cryptographic constructions beyond signature and encryption schemes. In this paper, we propose a more efficient module lattice version of Jiang et al.’s commitment schemes. Precisely, we construct all group-action-based components from their framework to the module lattice setting, thereby constructing commitment schemes that rely on specific instances of the module lattice isomorphism problem (module-LIP). Simultaneously, we investigate the hardness of these module-LIP problems to provide security foundations for the schemes, while extending the theoretical foundations of lattice isomorphism problems in the context of module lattices.

Keywords: module-LIP · Group action · Commitment

1 Introduction

The Lattice Isomorphism Problem (LIP) asks to determine whether two given lattices are isomorphic, i.e., whether there exists an orthogonal bijective transformation between them (decision version), or to find such a transformation (search version). Since Ducas and van Woerden [11] constructed cryptographic schemes based on LIP, including a key encapsulation mechanism and a signature scheme, a series of related constructions have been proposed [11, 10, 2, 4, 22]. Among them, the efficient signature scheme based on LIP, Hawk, was submitted to NIST and entered the second round of the call for additional signatures. Roughly speaking, the idea of these approaches is to use LIP to mask a known lattice’s favorable geometric structure, thereby allowing the constructor to leverage this structure while an adversary cannot do so unless the hardness assumption is broken.

Another natural approach is to view LIP as a matrix group acting on quadratic forms. In this direction, Jiang et al. proposed a universal framework for constructing commitment schemes using group actions and instantiated it with LIP [16]. Roughly speaking, the idea of their scheme is to consider a set equipped with a group action. One selects an element x as the public commitment key (CK), and uses $g \star x$ as the commitment to bind a group element g . To break the **binding** property, the committer would need to find a non-trivial element in the stabilizer subgroup of x , and this task is hard for certain group actions. To extract the message **hided** perfectly in $g \star x$ from g , a component called extractor is defined. Finally, to sample g in the group such that its distribution is uniform under the condition that $g \star x$ is fixed, they defined a component called re-randomization algorithm.

It is worth noting that, as mentioned earlier, when designing schemes using LIP, it is desirable that the specific lattice to be masked possesses sufficiently favorable geometric properties, while ensuring that it is sufficiently difficult for an adversary to remove this masking. Therefore, selecting a suitable specific lattice and analyzing the hardness of the corresponding LIP are crucial. Currently, the most prominent concerns are the LIP for the lattice $\mathbb{Z}^n$, i.e., $\mathbb{Z}$ -LIP [2,8], and its algebraically structured counterpart, the $\mathcal{O}_{\mathbb{L}}^2$ -LIP problem, where $\mathcal{O}_{\mathbb{L}}^2$ denotes the module lattice structure over a cyclotomic number field of conductor a power of two, denoted $\mathbb{L}$ [27,24,6,1]. Beyond these, the LIP corresponding to code-constructed lattices and Barnes-Wall (BW) lattices has also been studied [9,30,20].

Moreover, in both approaches, breaking the scheme typically requires solving the search version of LIP, whereas security reductions are usually constructed using the decision version. Unlike LWE, the search-to-decision reduction for LIP is far from trivial. However, it appears that beyond computing the genus invariant, there are no better algorithms for solving the decision LIP than solving the search LIP itself. Consequently, the current practice is to regard the decision LIP between two lattices as being as hard as determining whether they lie in the same genus. In recent years, the community has also attempted to introduce new invariants of a similar nature [23,25].

Thus, selecting a suitable pair of lattices within a given genus has become a focus. The computability of such a selection was addressed in [33,16]. However, corresponding results for structured lattices are still lacking. For the former approach, HAWK establishes its security proof by introducing a new hard problem (which, in a sense, has not yet been fully understood) to replace decision-LIP.

Previously, the only search-to-decision reduction for LIP was given in [15], which reduced $\mathbb{Z}$ -LIP to a set of decision LIPs. It was not until the recent work [14] that this reduction was extended to reduce the LIP corresponding to direct sum lattices over general rings to certain decision LIPs. An important example is the reduction of $\mathcal{O}_{\mathbb{L}}^2$ -LIP to a series of decision LIPs between module lattices generated by

$$B_k^{(0)} = \begin{pmatrix} 1 & 0 \\ 0 & (1+X)^k \end{pmatrix}, \quad B_k^{(1)} = \begin{pmatrix} 1 & 0 \\ (1+X)^{k-1} & (1+X)^k \end{pmatrix},$$

where X is a primitive root of unity in $\mathbb{L}$. This provides pairs of lattices for decision-LIP that inspire greater confidence than those merely lying in the same genus. Whether the lattices in one pair exhibit some intrinsic differences—whether geometric or algebraic—is a question that merits further study.

1.1 Our Contributions

Since algebraic structure usually brings more efficiency, in this paper, we instantiate all components involved in Jiang et al.’s framework [16] using the module version of LIP and investigate the hardness of the underlying problems.

To construct the components with module lattice, a naive idea is to replace the group $\text{GL}_n(\mathbb{Z})$ with $\text{GL}_2(\mathcal{O}_{\mathbb{L}})$, a group related to module lattice. However, obstacles arise when constructing the re-randomization algorithm, since unlike general lattices that can use efficient algorithms (e.g., LLL algorithm) to find an integral basis, there is currently no efficient algorithm on classical computers for finding a $\mathcal{O}_{\mathbb{L}}$ -basis of module lattices. This issue is also reflected in the self-reduction of the module lattice isomorphism problem. To bypass this difficulty, the work [10] considers the group $\text{SL}_2(\mathcal{O}_{\mathbb{L}})$. However, if we adopt $\text{SL}_2(\mathcal{O}_{\mathbb{L}})$ directly, we will face another obstacle when constructing deterministic extractors (since we can no longer use the determinant as the extractor as in [16] any more). To meet both requirements simultaneously, we have to exploit a new group action, the corresponding re-randomization algorithm and the extractor.

By introducing the Generalized Special Linear group $\text{GSL}_2(\mathcal{O}_{\mathbb{L}})$ over $\mathcal{O}_{\mathbb{L}}$ and its projective group $\text{PGSL}_2(\mathcal{O}_{\mathbb{L}})$, along with selecting appropriate isomorphism classes of Hermitian forms, we define a new group action and successfully design the required re-randomization algorithm together with the extractor.

We would like to point out that it is challenging to prove that our designed algorithm satisfies the requirements of a re-randomization algorithm. We have to carefully switch between the new group action and the original one to leverage the properties brought by the original self-reduction algorithm. In fact, this switch may only hold for the equivalence class of diagonal Hermitian forms.

Another challenge is to find a pair of Hermitian forms such that the extractor can uniformly sample all possible information from the automorphism group of one form, while only extracting fixed information from the automorphism group of the other. Moreover, the distinguishing problem corresponding to this pair of Hermitian forms must be computationally hard. Even for $\mathbb{Z}^n$, [16] just provides the existence of such a pair of Hermitian forms (more precisely the quadratic forms) and some possible ways to find them, instead of presenting the explicit description of the pair of Hermitian forms. However, in this paper, we successfully describe the desired pair of Hermitian forms by adjusting the concrete hard decision LIP instances $B_k^{(0)}, B_k^{(1)}$ presented in [14].

Furthermore, the security of our commitment schemes is directly based on a special decision version of LIP and a type of lattice automorphism problem (LAP) rather than the more standard search version of LIP as in [16]. Regarding the decision version of LIP, a recent work [14] presents a reduction from $\mathcal{O}_{\mathbb{L}}^2$ -LIP to a family of decision LIP problems, offering more promising concrete

hard decision LIP instances. Regarding LAP, a reduction from $\mathbb{Z}$ -LIP to $\mathbb{Z}$ -LAP, presented in [17], demonstrates the hardness of $\mathbb{Z}$ -LAP. For the module-LAP, we prove that it is at least as hard as that of $\mathcal{O}_{\mathbb{L}}^2$ -LIP, which is one of the underlying hard problems of HAWK. Although a series of works [27, 24, 6, 1] have attempted to study the hardness of $\mathcal{O}_{\mathbb{L}}^2$ -LIP, there has been no substantial progress in the case where $\mathbb{L}$ is a cyclotomic field.

In general, reductions on structured lattices typically require more complex arguments than reductions on generic lattice problems. Benefiting from the simple structure brought by rank 2, we can adopt the technique of decomposing module lattices via eigenspaces, as used in [24], to achieve the reduction. The main challenge for the reduction lies in the fact that the eigenvalues of automorphisms may not lie within the originally considered field, making it impossible to compute eigenspaces over it. To overcome this, we need to operate over a field extension that includes these eigenvalues. To do these operations requires us to prove that the extended field remains a CM field and that the ring of integers over this extended field can be efficiently computed. For the cyclotomic fields, which are of our primary interest, there already exist abundant results. We also leverage some techniques from algebraic number theory to extend these results to the general case.

Overall, with these components and the framework in [16], we can obtain a series of commitment schemes. Our approach leverages the structured properties of module lattices to achieve improved efficiency, and the security guarantee is based on the hardness of $\mathcal{O}_{\mathbb{L}}^2$ -LIP and certain decision module-LIPs. The primary contributions of our work are summarized as follows:

- We identify a suitable group action in the module lattice setting and develop a re-randomization algorithm and an extractor on it, resulting in a series of commitment schemes based on module-LIP within the framework of [16].
- We investigate the hardness assumptions underlying our schemes to provide security foundations, while extending the theoretical foundations of lattice isomorphism problems in the context of module lattices.

It should be noted that although our instantiation on structured lattices offers significant performance improvements over the instantiation on $\mathbb{Z}^n$ lattices in [16], it is still far from a practical scheme. In particular, the generation of commitments involves substantial computational overhead due to the high complexity of the computation of Hermite normal form (invoked in HermiteSolve). In practical implementations, we aim to leverage algebraic structure optimizations such as **TowerSolve** [28], though this would compromise the security reduction in the original framework. We briefly discuss this issue at the end of the article.

Roadmap. The remainder of this paper is organized as follows. Sec. 2 provides the necessary background and preliminaries. Sec. 3 presents some hardness reductions for module-LIP and module-LAP over certain module lattices related to $\mathcal{O}_{\mathbb{L}}^2$ -LIP, including the reduction from $\mathcal{O}_{\mathbb{L}}^2$ -LIP to $\mathcal{O}_{\mathbb{L}}^2$ -LAP. These reductions will serve as the security foundations for the commitment schemes constructed

in the next section. Sec. 4 presents the components required by the construction framework, subsequently obtains the commitment schemes, and concludes with a brief analysis of the schemes' complexity. Finally, Sec. 5 concludes the paper.

2 Notations and preliminaries

2.1 Notations

- We use $x \leftarrow \mathcal{D}$ to denote that x is sampled from a distribution $\mathcal{D}$. In this paper, we focus solely on discrete distributions. For a finite set S , we denote by $U(S)$ the uniform distribution over S , and write $s \leftarrow_{\S} S$ to indicate that s is drawn uniformly from S .
- The Euclidean norm of $a \in \mathbb{R}^n$ is denoted by $\|a\|$. Let $GL_n(\mathbb{R})$ and $GL_n(\mathbb{Z})$ be the general linear group of rank n over $\mathbb{R}$ and $\mathbb{Z}$, respectively.
- We use J_2 to represent the matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, J_B to represent $B^{-1}J_2B$ for some 2×2 matrix B . We use rI_n to represent the matrix $\text{diag}(r, r, \dots, r)$.
- For a number field $\mathbb{K}$, the parameter $\mathbf{K}$ denotes degree of $\mathbb{K}$, $\log \Delta_{\mathbb{K}}$, and a basis of $\mathcal{O}_{\mathbb{K}}$.
- For x in a number field $\mathbb{K}$, we call x^* is the complex conjugation of x if $\sigma(x^*) = \overline{\sigma(x)}$, $\forall \sigma \in \text{Hom}_{\mathbb{Q}}(\mathbb{K}, \mathbb{C})$. For matrix $H = (h_{ij}) \in \mathbb{K}^{m \times n}$, let H^* denote $(h_{ij}^*)^T$ and $\overline{H}$ denote (h_{ij}^*) if all h_{ij}^* exist.
- $\mathcal{S}_n^{>0}(\mathbb{R})$ consists of all positive definite quadratic forms over $\mathbb{R}$. For a CM number field $\mathbb{L}$, the Hermitian forms $\mathcal{H}_{\ell}^{>0}(\mathbb{L})$ consists of all $\mathbf{Q} \in \mathbb{L}^{\ell \times \ell}$ such that $\mathbf{Q}^* = \mathbf{Q}$ and $\text{Tr}_{\mathbb{L}}(v^* \mathbf{Q} v) > 0$ for all $v \in \mathbb{L}^{\ell} \setminus \{0\}$.
- For a ring A in a number field that are closed under complex conjugating, the unitary matrices over A is $\mathcal{U}_n(A) := \{T \in M_n(A) | T^* T = I_n\}$.
- For a number field $\mathbb{K}$, we use $\mu(\mathbb{K})$ to denote the roots of unity in $\mathbb{F}$. Note $\mu(\mathbb{K}) \subset \mathcal{O}_{\mathbb{K}}$ and $\mu(\mathbb{K}) = \mathcal{U}_1(\mathcal{O}_{\mathbb{K}})$.
- For a number field $\mathbb{L}$, we use $(\mathcal{O}_{\mathbb{L}}^2)^*$ to denote the vector $\begin{pmatrix} f \\ g \end{pmatrix}$ in $\mathcal{O}_{\mathbb{L}}^2$ such that $(f) + (g) = \mathcal{O}_{\mathbb{L}}$.
- For $\mathbb{L} = \mathbb{Q}[X]/(\Phi_{2^d}(X)) = \mathbb{Q}[X]/(X^{2^{d-1}} + 1)$, let $\Pi = 1 + X$, $\mathbf{p} = \langle \Pi \rangle$. Define $B_k^{(0)} = \begin{pmatrix} 1 & 0 \\ 0 & \Pi^k \end{pmatrix}$, $B_k^{(1)} = \begin{pmatrix} 1 & 0 \\ \Pi^{k-1} & \Pi^k \end{pmatrix}$, $B_k^{(2)} = \begin{pmatrix} 1 & 0 \\ (1 + X^2)^{k-1} & (1 + X^2)^k \end{pmatrix}$, and $\mathbf{Q}_k^{(i)} = (B_k^{(i)})^* B_k^{(i)}$, $i = 0, 1, 2$.
- $\lceil \cdot \rceil : \mathbb{R} \rightarrow \mathbb{Z}$ denotes the rounding function.

2.2 Lattices and quadratic forms

A lattice is a discrete additive subgroup of $\mathbb{R}^m$ (or less commonly $\mathbb{C}^m$). Typically, one constructs a lattice by selecting n linearly independent vectors $b_1, b_2, \dots, b_n$

in $\mathbb{R}^m$, so that every lattice element can be expressed as an integer linear combination of these basis vectors. In other words, if we denote the basis by $B = (b_1, \dots, b_n)$, then the lattice $\mathcal{L}(B)$ is given by $\{Bz : z \in \mathbb{Z}^n\}$, which defines a lattice of rank n in $\mathbb{R}^m$. Denote the i -th minima of $\mathcal{L}$ by $\lambda_i(\mathcal{L})$.

The inner product with respect to $\mathbf{Q} \in \mathcal{S}_n^{>0}(\mathbb{R})$ is defined as $\langle x, y \rangle_{\mathbf{Q}} : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$, where $(x, y) \mapsto x^T \cdot \mathbf{Q} \cdot y$. The norm with respect to $\mathbf{Q} \in \mathcal{S}_n^{>0}(\mathbb{R})$ is defined as $\|x\|_{\mathbf{Q}} = \sqrt{\langle x, x \rangle_{\mathbf{Q}}}$. Note that for a basis B and vectors $x, y \in \mathbb{R}^n$, we have $\langle Bx, By \rangle = x^T B^T B y = \langle x, y \rangle_{B^T B}$. Thus, the geometry of the lattice $\mathcal{L}(B)$ is fully described by $\mathbf{Q} = B^T B$. We can define $\lambda_i(\mathbf{Q})$ as $\lambda_i(\mathcal{L}(B))$.

2.3 Discrete gaussian sampling

Let σ be a positive real number. We define the Gaussian mass $\rho_\sigma : \mathbb{R}^n \rightarrow \mathbb{R}$ by $\rho_\sigma(x) = \exp(-\|x\|^2/\sigma^2)$ for any $x \in \mathbb{R}^n$. The discrete Gaussian distribution $D_{\mathcal{L}, \sigma}$ over the lattice $\mathcal{L}$ assigns probability $\rho_\sigma(x)/\sum_{y \in \mathcal{L}} \rho_\sigma(y)$ to each $x \in \mathcal{L}$, and zero otherwise.

We also define the Gaussian mass with respect to a quadratic form $\mathbf{Q} \in \mathcal{S}_n^{>0}(\mathbb{R})$ as $\rho_{\mathbf{Q}, \sigma} : \mathbb{R}^n \rightarrow \mathbb{R}$, given by $\rho_{\mathbf{Q}, \sigma}(x) = \exp(-\|x\|_{\mathbf{Q}}^2/\sigma^2)$ for any $x \in \mathbb{R}^n$. If $\mathbf{Q} = B^T B$ for some basis B of $\mathcal{L}$, then $D_{\mathbf{Q}, \mathbb{Z}^n, \sigma}(x)$ is equivalent to $D_{\mathcal{L}, \sigma}(B \cdot x)$, as $\rho_{\mathbf{Q}, \sigma}(x) = \rho_\sigma(B \cdot x)$.

When σ is sufficiently large compared to the maximum length of the Gram-Schmidt basis vectors, efficient sampling from the discrete Gaussian distribution is possible. The following lemma provides the conditions under which this is achievable.

Lemma 2.1 ([10, Lemma 1]) *There exists a probabilistic polynomial-time algorithm that, given a quadratic form $\mathbf{Q} \in \mathcal{S}_n^{>0}(\mathbb{R})$, and a parameter $\sigma \geq \|\tilde{B}_{\mathbf{Q}}\|/\pi \cdot \sqrt{\log(2n+4)}$, outputs a sample according to the distribution $D_{\mathbf{Q}, \mathbb{Z}^n, \sigma}$. Here $B_{\mathbf{Q}} \in M_n(\mathbb{R})$ such that $\mathbf{Q} = B_{\mathbf{Q}}^T B_{\mathbf{Q}}$ and $\tilde{B}_{\mathbf{Q}}$ denotes the Gram-Schmidt orthogonalization of $B_{\mathbf{Q}}$, and $\|\tilde{B}_{\mathbf{Q}}\|$ is the length of the longest vector in it. (This is independent of the specific decomposition of $\mathbf{Q}$.)*

Further more, by first LLL reducing $\mathbf{Q}$, we can replace the condition on σ with $\sigma \geq 2^{\Theta(n)} \lambda_n(\mathbf{Q})$.

2.4 Number Theory

A number field $\mathbb{K}$ is a finite extension of the rational numbers $\mathbb{Q}$. Equivalently, any such field may be expressed as $\mathbb{Q}[X]/(P)$, where P is a monic irreducible polynomial whose degree matches that of the extension. For a field $\mathbb{K}$ of degree d , there exist exactly d embeddings $\sigma_1, \dots, \sigma_d$ into $\mathbb{C}$. Those mappings that send $\mathbb{K}$ into $\mathbb{R}$ are called real embeddings, while the others, which occur in complex conjugate pairs, are referred to as complex embeddings. If we denote by r_1 the number of real embeddings and by r_2 the number of pairs of complex embeddings, then $d = r_1 + 2r_2$. A field is said to be totally real if $r_1 = d$, and totally imaginary if $2r_2 = d$.

Canonical embedding. The **canonical embedding** of a number field $\mathbb{K}$ is defined by the mapping $\Sigma_{\mathbb{K}} : x \mapsto (\sigma_1(x), \dots, \sigma_d(x))^T$, which sends elements into $\mathbb{C}^d$. Often, one identifies $\mathbb{K}$ with its image under this mapping so that the ring of integers $\mathcal{O}_{\mathbb{K}}$ acquires a lattice structure in $\mathbb{C}^d$. It is important to note, we are not representing elements in $\mathbb{K}$ using the canonical embedding. The norm on $\mathbb{K}$ is defined as $\mathcal{N}_{\mathbb{K}}(z) = \prod_{i=1}^d \sigma_i(z)$ and the trace as $\text{Tr}_{\mathbb{K}}(z) = \sum_{i=1}^d \sigma_i(z)$. By considering the $\mathbb{Q}$ -linear map $m_z : x \mapsto zx$, one sees that $\mathcal{N}_{\mathbb{K}}(z)$ equals the determinant of m_z and $\text{Tr}_{\mathbb{K}}(z)$ equals its trace; notably, both quantities belong to $\mathbb{Q}$.

We also have the **coefficient embedding** $\text{vec} : \mathbb{Q}[X]/(P) \rightarrow \mathbb{Q}^n$, $a_0 + a_1X + \dots + a_{n-1}X^{n-1} \mapsto (a_0, a_1, \dots, a_{n-1})^T$, which is an additive group isomorphism. When P is a cyclotomic polynomial of 2-power order, i.e., when $\mathbb{Q}[X]/(P)$ is a cyclotomic field of 2-power order, the canonical embedding and the coefficient embedding differ only by a scaling factor geometrically.

CM number field. A number field $\mathbb{L}$ is a CM number field if it is a quadratic extension $\mathbb{L}/\mathbb{K}$ where the base field $\mathbb{K}$ is totally real and $\mathbb{L}$ is totally imaginary. An equivalent condition is $\mathbb{L} = \mathbb{K}(\alpha)$ with $\mathbb{K}$ totally real, $\alpha^2 \in \mathbb{K}$ and $\phi(\alpha^2) < 0$ for any homomorphism $\phi : \mathbb{K} \rightarrow \mathbb{C}$. The extension $\mathbb{L}/\mathbb{K}$ is a Galois extension and we denote the Galois group by $\text{Gal}(\mathbb{L}/\mathbb{K})$. There is a complex conjugation in $\text{Gal}(\mathbb{L}/\mathbb{K})$, i.e. existing $\tau \in \text{Gal}(\mathbb{L}/\mathbb{K})$ s.t. $\forall x \in \mathbb{L}, \sigma_i(\tau(x)) = \overline{\sigma_i(x)}$. We usually denote $\tau(x)$ by x^* . Then the natural Hermitian inner product $\langle \Sigma(x), \Sigma(y) \rangle$ is equal to $\text{Tr}_{\mathbb{L}}(x^*y)$. As an important example, the cyclotomic number fields are all CM number fields.

Rings of integer and ideals. Let $\mathcal{O}_{\mathbb{L}}$ denote the ring of integers of a number field $\mathbb{L}$. $\mathcal{O}_{\mathbb{L}}$ is a free $\mathbb{Z}$ -module of rank d . The discriminant of $\mathbb{L}$, denoted $\Delta_{\mathbb{L}}$, is defined as $(\det(\sigma_i(\alpha_j)))_{i,j}^2 \in \mathbb{Z}$, where $(\alpha_j)_{1 \leq j \leq d}$ is any basis of $\mathcal{O}_{\mathbb{L}}$. Specifically, there exists some absolute constant $c > 1$ such that $\Delta_{\mathbb{L}} \geq c^d$ for all number fields $\mathbb{L}$ (due to Minkowski [29, pp. 261–264]). In particular, we always have $d = \text{poly}(\log \Delta_{\mathbb{L}})$. An (fractional) ideal $\mathcal{I}$ is an finitely generated additive subgroup of $\mathbb{L}$ such that $x \cdot \mathcal{I} \subseteq \mathcal{I}$ for all $x \in \mathcal{O}_{\mathbb{L}}$. When an ideal is contained in $\mathcal{O}_{\mathbb{L}}$, we call it an integral ideal. We use $(x) := x \cdot \mathcal{O}_{\mathbb{L}}$ to denote the principal ideal generated by a single element $x \in \mathbb{L}$.

Module lattices and Hermitian forms. Assume $\mathbb{L}$ is a CM number field with degree d . A module lattice M (over $\mathbb{L}$) is an $\mathcal{O}_{\mathbb{L}}$ module in $\mathbb{L}^{\ell}$. Through canonical embedding $\Sigma_{\mathbb{L}}$, we can map M to a lattice in $\mathbb{C}^{d\ell}$. It has the form $b_1\mathcal{I}_1 + \dots + b_r\mathcal{I}_r$, where the $\mathcal{I}_i$'s are non-zero fractional ideals of $\mathbb{L}$ and $(b_1, \dots, b_r)$ are $\mathbb{L}$ -linearly independent vectors of $\mathbb{L}^{\ell}$, for some $\ell > 0$. We call $\mathbf{B} = (B, (\mathcal{I}_i)_{1 \leq i \leq r})$ a pseudo-basis for M , where B is the matrix whose columns are the b_i . If there is a pseudo-basis of M s.t. all $\mathcal{I}_i$ is $\mathcal{O}_{\mathbb{L}}$, we call M a free module and B the free basis of M .

When $\mathbb{L}$ is a CM number field, the natural Hermitian inner product of $v, w \in \mathbb{L}^{\ell}$ under the embedding is equal to $\text{Tr}_{\mathbb{L}}(v^* \cdot w)$. Similar to the definition over lattices, the inner product with respect to $\mathbf{Q} \in \mathcal{H}_{\ell}^{>0}(\mathbb{L})$ is defined as $\langle x, y \rangle_{\mathbf{Q}} : \mathbb{L}^{\ell} \times \mathbb{L}^{\ell} \rightarrow \mathbb{Q}$, where $(x, y) \mapsto \text{Tr}(x^* \cdot \mathbf{Q} \cdot y)$. The norm with respect to $\mathbf{Q} \in \mathcal{H}_{\ell}^{>0}(\mathbb{L})$

is defined as $\|x\|_{\mathbf{Q}} = \sqrt{\langle x, x \rangle_{\mathbf{Q}}}$. We omit the subscript $\mathbf{Q}$ when $\mathbf{Q} = I_{\ell}$. Note that for a $B \in \mathbb{L}^{\ell \times \ell}$ and vectors $x, y \in \mathbb{L}^{\ell}$, we have $\langle Bx, By \rangle = \langle x, y \rangle_{B^*B}$.

So we can define the Gaussian mass $\rho_{\sigma} : \mathbb{L}^{\ell} \rightarrow \mathbb{R}$ by $\rho_{\sigma}(v) = \exp(-\text{Tr}_{\mathbb{L}}(v^*v)/\sigma^2)$ for any $v \in \mathbb{L}^{\ell}$ and the Gaussian mass with respect to a Hermitian form $\mathbf{Q} \in \mathcal{H}_{\ell}^{>0}(\mathbb{L})$ as $\rho_{\mathbf{Q},\sigma} : \mathbb{L}^{\ell} \rightarrow \mathbb{R}$, given by $\rho_{\mathbf{Q},\sigma}(x) = \exp(-\|x\|_{\mathbf{Q}}^2/\sigma^2)$ for any $x \in \mathbb{L}^{\ell}$.

Cases on HAWK's setting. In HAWK, they consider $\mathbb{L} = \mathbb{Q}[X]/(\Phi_{2^d}(X)) = \mathbb{Q}[X]/(X^{2^{d-1}} + 1)$ to be a power of two cyclotomic number field and $\zeta_{2^d} := X$ to be a 2^d -th root of unity. Let $n = 2^{d-1} = \deg(\mathbb{L})$, $\Pi = 1 + X$, $\mathfrak{p} = \langle \Pi \rangle$. We have $\mathfrak{p}^{2^k} = \langle 1 + X^{2^k} \rangle$ for any $0 \leq k \leq d-2$ and odd j , and $\langle 2 \rangle = \mathfrak{p}^n \subset \mathfrak{p}$. A corollary is that $\overline{\mathfrak{p}^s} = \mathfrak{p}^s = \mathfrak{p}^s$ for any $s \in \mathbb{N}$.

In this case, we have $\mathcal{O}_{\mathbb{L}} = \mathbb{Z}[X]/(X^{2^{d-1}} + 1)$, and we directly represent elements in the field using their polynomial forms. This can also be understood as choosing $\{1, X, \dots, X^{n-1}\}$ as a basis of $\mathcal{O}_{\mathbb{L}}$, where the resulting vector representation corresponds to the coefficient embedding. Under the coefficient embedding

$$\text{vec} : a_0 + a_1X + \dots + a_{n-1}X^{n-1} \mapsto \begin{pmatrix} a_0 \\ \vdots \\ a_{n-1} \end{pmatrix}, \quad \mathcal{O}_{\mathbb{L}} \text{ corresponds to } \mathbb{Z}^n. \text{ Under this}$$

basis, the linear transformation m_f corresponding to multiplication by f is represented by the matrix

$$\text{rot}(f) = (\text{vec}(f) \mid \text{vec}(Xf) \mid \dots \mid \text{vec}(X^{n-1}f)) \in \mathbb{Q}^{n \times n}, \quad \text{for } f \in \mathbb{L},$$

and this naturally extends to matrices $B \in \mathbb{L}^{k \times \ell}$ as

$$\text{rot}(B) = \begin{pmatrix} \text{rot}(B_{11}) & \dots & \text{rot}(B_{1\ell}) \\ \vdots & & \vdots \\ \text{rot}(B_{k1}) & \dots & \text{rot}(B_{k\ell}) \end{pmatrix} \in \mathbb{Q}^{nk \times n\ell}.$$

We have $\|f\|^2 = n\|\text{vec}(f)\|^2$ and so $\rho_{\mathbf{Q},\sigma}(f/\sqrt{n}) = \rho_{\mathbf{Q},\sqrt{n}\sigma}(f) = \rho_{\text{rot}(\mathbf{Q}),\sigma}(\text{vec}(f))$.

The conjugation and multiplication-by- X operations are then expressed as $\tau : a_0 + a_1X + \dots + a_{n-1}X^{n-1} \mapsto a_0 - (a_{n-1}X + \dots + a_1X^{n-1})$, and $m_X : a_0 + a_1X + \dots + a_{n-1}X^{n-1} \mapsto -a_{n-1} + a_0X + \dots + a_{n-2}X^{n-1}$. They both belong to $\text{Aut}(\mathbb{Z}^n) = S_n^{\pm}$.

In the setting considered by HAWK, it is required to sample matrices in SL_2 over the 2-power cyclotomic ring $\mathcal{O}_{\mathbb{L}}$ both in its key generation and in the self-reduction described in the paper. The approach is to first perform a sampling over $\mathcal{O}_{\mathbb{L}}^2$, and then complete it into a matrix in SL_2 . The following lemma guarantees the existence of an efficient algorithm for the latter step.

Lemma 2.2 ([10, Algorithm 4]) *There is an efficient deterministic algorithm HermiteSolve that input the field $\mathbb{L}$ and $(f, g) \in \mathcal{O}_{\mathbb{L}}^2$, output $(F, G) \in \mathcal{O}_{\mathbb{L}}^2$ s.t. $fG - FG = 1$ when $(f) + (g) = \mathcal{O}_{\mathbb{L}}$ and $\perp$ otherwise.*

The following Heuristic Assumption guarantees that when the first step of sampling is performed via discrete Gaussian sampling, the probability that the

above algorithm fails is approximately $1/4$. In Appendix A of the full version [12], they give a heuristic explanation and matching experimental evidence for $\mathbf{Q} = I_2$. Indeed, for this case, Lemma 4.4 in [31] establishes a constant lower bound. For $\mathbf{Q} = \mathbf{Q}_k^{(0)}$, our experimental verification for small k also aligns with this heuristic expectation.

Heuristic 1 (For $\mathbf{Q}$) Suppose $n \geq 4$ and $\sigma \geq \Omega(n)$, the probability value

$$\rho_{\mathbf{Q},\sigma}((\mathcal{O}_{\mathbb{L}}^2)^*)/\rho_{\mathbf{Q},\sigma}(\mathcal{O}_{\mathbb{L}}^2)$$

is around $\frac{3}{4}$.

A simple observation is that for any $U \in \text{GL}_2(\mathcal{O}_{\mathbb{L}})$, we have $U \cdot \mathcal{O}_{\mathbb{L}}^2 = \mathcal{O}_{\mathbb{L}}^2$ and $U \cdot (\mathcal{O}_{\mathbb{L}}^2)^* = (\mathcal{O}_{\mathbb{L}}^2)^*$. Hence, $\rho_{U^*\mathbf{Q}U,\sigma}(\mathcal{O}_{\mathbb{L}}^2) = \rho_{\mathbf{Q},\sigma}(\mathcal{O}_{\mathbb{L}}^2)$ and $\rho_{U^*\mathbf{Q}U,\sigma}((\mathcal{O}_{\mathbb{L}}^2)^*) = \rho_{\mathbf{Q},\sigma}((\mathcal{O}_{\mathbb{L}}^2)^*)$.

2.5 Module-LIP, module-LAP and module- Δ LIP

For simplicity, let $\mathbb{L}$ be a CM number field in this subsection.

Definition 2.1 Two Hermitian forms $\mathbf{Q}_0$ and $\mathbf{Q}_1$ over $\mathbb{L}$ are said to be isomorphic if there exists $U \in \text{SL}_n(\mathcal{O}_{\mathbb{L}})$ (resp. $\text{GL}_n(\mathcal{O}_{\mathbb{L}})$) such that $\mathbf{Q}_1 = U^*\mathbf{Q}_0U$. The isomorphism class of $\mathbf{Q}_0$ is denoted by $[\mathbf{Q}_0]_{\text{sl}}$ (resp. $[\mathbf{Q}_0]_{\text{gl}}$). Sometimes we write $[\mathbf{Q}_0]_{\text{sl}}$ as $[\mathbf{Q}_0]$ for simplicity.

Definition 2.2 (Module-Lattice isomorphism Problem, module-LIP) Given Hermitian forms $\mathbf{Q}_1 \in [\mathbf{Q}_0]$ and parameters of $\mathbb{L}$, the objective is to find an $U \in \text{SL}_{\ell}(\mathcal{O}_{\mathbb{L}})$ such that $\mathbf{Q}_1 = U^*\mathbf{Q}_0U$. We denote this problem as module-LIP $_{\mathbb{L},\mathbf{Q}_0,\mathbf{Q}_1}$, and typically omit the subscript $\mathbb{L}$ when no ambiguity arises. If $\mathbf{Q}_0$ is fixed, denote by

- module-LIP $_{[\mathbf{Q}_0]}$ the problem of solving module-LIP $_{\mathbf{Q}_0,\mathbf{Q}_1}$ for any $\mathbf{Q}_1 \in [\mathbf{Q}_0]$,
- module-LIP $_{\mathcal{D}_{[\mathbf{Q}_0]}}$ the problem of solving module-LIP $_{\mathbf{Q}_0,\mathbf{Q}_1}$ for $\mathbf{Q}_1$ sampled from the distribution $\mathcal{D}_{[\mathbf{Q}_0]}$.

Remark 1. The choice of the group for the transition matrix U induces different isomorphism classes. Note that for $U \in \text{GL}_n(\mathcal{O}_{\mathbb{L}})$, given $\mathbf{Q} = U^*\mathbf{Q}_0U$ in $[\mathbf{Q}_0]_{\text{gl}}$, we can first apply the algorithm LS1 (as in Proposition 2.2) to $\det(\mathbf{Q})/\det(\mathbf{Q}_0)$ to compute $\det(U)$, and then perform a congruence transformation on $\mathbf{Q}$ using $\text{diag}(\det(U)^{-1}, 1, \dots, 1)$ to convert it into an instance in $[\mathbf{Q}_0]_{\text{sl}}$. Therefore, the choice of isomorphism class does not affect the hardness of the problem. The same reasoning applies to the decision version of module-LIP.

Definition 2.3 For an hermitian form $\mathbf{Q}$ over $\mathbb{L}$, we define the automorphism group of it by $\{U \in \text{GL}_{\ell}(\mathcal{O}_{\mathbb{L}}) | U^*\mathbf{Q}U = \mathbf{Q}\} \triangleq \text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q})$. Note that $\mu(\mathbb{L})I_{\ell} \subseteq \text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q})$ always holds; we refer to these automorphisms as trivial.

Remark 2. The automorphism group has such a property: $\text{Aut}_{\mathcal{O}_{\mathbb{L}}}(U^* \mathbf{Q} U) = U^{-1} \text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q})$. Indeed, we always have $\text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q}) \subseteq \text{GSL}_{\ell}(\mathcal{O}_{\mathbb{L}})$ (see its definition in [Definition 4.1](#)).

Definition 2.4 (module-Lattice Automorphism Problem, module-LAP)

Given an hermitian form $\mathbf{Q}$, the objective is to find a non-trivial module lattice automorphism $P \in \text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q}) \setminus \mu(\mathbb{L}) I_{\ell}$. We denote this problem as module-LAP $_{\mathbb{L}, \mathbf{Q}}$, and typically omit the subscript $\mathbb{L}$ when no ambiguity arises. Similarly to module-LIP, we can define module-LAP $_{[\mathbf{Q}]_{\text{gl}}}$ and module-LAP $_{\mathcal{D}_{[\mathbf{Q}]_{\text{gl}}}}$.

Example 2.1 Currently, the most prominent case is $\ell = 2$, $\mathbf{Q} = I_2$, corresponding to the hardness assumption used in HAWK. We denote these problems as $\mathcal{O}_{\mathbb{L}}^2$ -LIP and $\mathcal{O}_{\mathbb{L}}^2$ -LAP. And $\text{Aut}_{\mathcal{O}_{\mathbb{L}}}(I_2) = \mathcal{U}_2(\mathcal{O}_{\mathbb{L}})$.

Lemma 2.3 ([24, Lemma 3.1]) Let $\mathbb{L}$ be a CM number field with degree $2d$. Then

$$\mathcal{U}_2(\mathcal{O}_{\mathbb{L}}) = \left\{ \begin{pmatrix} \xi_1 & 0 \\ 0 & \xi_2 \end{pmatrix} \mid \xi_1, \xi_2 \in \mu(\mathbb{L}) \right\} \cup \left\{ \begin{pmatrix} 0 & \xi_1 \\ \xi_2 & 0 \end{pmatrix} \mid \xi_1, \xi_2 \in \mu(\mathbb{L}) \right\}.$$

Furthermore, $\#(\mathcal{U}_2(\mathcal{O}_{\mathbb{L}})) \leq 2\#(\mu(\mathbb{L}))^2 \leq 128d^4$.

Definition 2.5 (module- Δ LIP) For Hermitian forms $\mathbf{Q}_1$ and $\mathbf{Q}_0$, given $\mathbf{Q} \in [\mathbf{Q}_1] \cup [\mathbf{Q}_0]$ and parameters of $\mathbb{L}$, the objective is to determine whether $\mathbf{Q}$ belongs to $[\mathbf{Q}_0]$ or to $[\mathbf{Q}_1]$. We denote this problem as module- Δ LIP $_{\mathbb{L}, \mathbf{Q}, \mathbf{Q}_0, \mathbf{Q}_1}$, and typically omit the subscript $\mathbb{L}$ when no ambiguity arises. Denote by

- module- Δ LIP $_{[\mathbf{Q}_0], [\mathbf{Q}_1]}$ the problem of solving module- Δ LIP $_{\mathbf{Q}, \mathbf{Q}_0, \mathbf{Q}_1}$ for any $\mathbf{Q} \in [\mathbf{Q}_1] \cup [\mathbf{Q}_0]$,
- module- Δ LIP $_{\mathcal{D}_{[\mathbf{Q}_0]}, \mathcal{D}_{[\mathbf{Q}_1]}}$ the problem of solving module- Δ LIP $_{\mathbf{Q}, \mathbf{Q}_0, \mathbf{Q}_1}$ for $\mathbf{Q}$ sampled from the distribution $\mathcal{D}_{[\mathbf{Q}_0]}$ or the $\mathcal{D}_{[\mathbf{Q}_1]}$.

The following theorem is an important example under the general result in [14]. It presents a family of decision LIP and proves that if this family of decision LIP can be solved, then the $\mathcal{O}_{\mathbb{L}}^2$ -LIP can be solved. The original text uses the geometric version of the module-LIP representation, which is here adapted to the representation of the Hermitian form under a free basis. The core reduction technique remains the same, but for completeness, its proof will be provided in the [Appendix C](#).

Theorem 2.1 (Adapted from [14, Theorem 10.3]) In line with HAWK's configuration, let $B_k^{(0)} = \begin{pmatrix} 1 & 0 \\ 0 & \Pi^k \end{pmatrix}$ and $B_k^{(1)} = \begin{pmatrix} 1 & 0 \\ \Pi^{k-1} & \Pi^k \end{pmatrix}$, and $\mathbf{Q}_k^{(0)} = (B_k^{(0)})^* B_k^{(0)}$, $\mathbf{Q}_k^{(1)} = (B_k^{(1)})^* B_k^{(1)}$. The parameters of module- Δ LIP $_{[\mathbf{Q}_k^{(0)}], [\mathbf{Q}_k^{(1)}]}$ depend only on k and the field $\mathbb{L}$. For convenience, we denote it as $\Delta_{k, \mathbb{L}}$. There is a (quantum) reduction from $\mathcal{O}_{\mathbb{L}}^2$ -LIP to $\{\Delta_{k, \mathbb{L}}, k = 2, \dots, 2n^2\}$. And in the reduction, each solver for $\Delta_{k, \mathbb{L}}$ is invoked at most once.

Remark 3. Although the original paper mentions that the first $\log n + 1$ $\Delta_{k,\mathbb{L}}$ oracles can be omitted, this appears to be achieved through a form of guessing rather than fundamentally eliminating the need for these oracles. In fact, since each oracle is invoked only once and a simple guess yields the correct oracle output with probability $1/2$, we can always eliminate $O(\log n)$ oracles via guessing.

2.6 Algorithmic consideration

Representation of ideals and modules. Assume $B_{\mathcal{O}_{\mathbb{L}}} = (\alpha_j)_{j=1,\dots,d}$ is a basis of $\mathcal{O}_{\mathbb{L}}$. We represent elements in $\mathbb{L}$ by their coordinates in the basis $B_{\mathcal{O}_{\mathbb{L}}}$, which is a vector in $\mathbb{Q}^d$. For $x \in \mathbb{L}$ represented by the vector $(x_1, \dots, x_d)^T \in \mathbb{Q}^d$, we define $\text{size}(x) := \sum_i \text{size}(x_i)$, where $\text{size}(a/b) := \lceil \log_2 |a| \rceil + \lceil \log_2 |b| \rceil$ for $a, b \in \mathbb{Z}$ coprime. As is customary, we assume that in this paper the $B_{\mathcal{O}_{\mathbb{L}}}$ is always a LLL-reduced basis of $\mathcal{O}_{\mathbb{L}}$, i.e. $\sigma(B_{\mathcal{O}_{\mathbb{L}}})$ forms an LLL-reduced basis of $\mathcal{O}_{\mathbb{L}}$. This choice is made to ensure that the coefficients of $\alpha_i \alpha_j$ under $B_{\mathcal{O}_{\mathbb{L}}}$ representation do not blow up.

Basic algorithms. Here, we list some algorithms that will be used in the reduction presented in Section 3. The following lemma guarantees that the computation of roots of unity in a given field is efficient.

Lemma 2.4 ([26, Corollary 2.11]) *Let $\mathbb{K}$ be a degree d number field. Then, $\mathbb{K}$ has at most $2d^2$ roots of unity, and there is a polynomial-time algorithm that, given a basis of the ring of integers $\mathcal{O}_{\mathbb{K}}$, computes the roots of unity in $\mathbb{K}$.*

The following lemma guarantees that we can compute the intersection of a module lattice and a $\mathbb{L}$ -linear space.

Lemma 2.5 ([24, Lemma 4.4]) *Let $\mathbb{L}$ be a number field, and $B_{\mathcal{O}_{\mathbb{L}}}$ be a $\mathbb{Z}$ basis of $\mathcal{O}_{\mathbb{L}}$. Then for $A \in \mathbb{L}^{2 \times 2}$ and a lattice $\mathcal{L} \subseteq \mathbb{L}^2$, there is a deterministic polynomial-time algorithm that, given $B_{\mathcal{O}_{\mathbb{L}}}$, A , and a basis $B_{\mathcal{L}}$ of $\mathcal{L}$, outputs $\ker(A) \cap \mathcal{L}$.*

Proposition 2.1 ([24, Proposition 3.2]) *Let $B \in GL_2(\mathbb{L})$, and $r \in \mathbb{L}$. Define $t_* : \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{L}^2 \mapsto \begin{pmatrix} x^* \\ y^* \end{pmatrix} \in \mathbb{L}^2$. It's a $\mathbb{Q}$ linear map. Given as input a basis of $\mathcal{O}_{\mathbb{L}}$, $G = B^* B$, and $\det(B)$, we can compute $B^{-1} J_2 t_* B$ and $m_r := B^{-1}(r I_2) B$ in polynomial time in the input size.*

From a constructive perspective, we will use the Hermitian form version of the free module-LIP as the hardness assumption. However, in some reductions, certain geometric versions of the problem are more convenient, because we need to perform operations such as intersection and summation, which can disrupt the free module representation (for example, the original version of Theorem 2.1 in [14]). In plain LIP, the conversion between the geometric and quadratic form versions is straightforward.

For module-LIP, the first challenge is to recover a free basis from a pseudo-basis of a free module, which is non-trivial. The folklore approach involves first computing an almost-free basis and then generating a generator for the remaining principal ideal of $\mathcal{O}_{\mathbb{L}}$, which can be done in polynomial time on a quantum computer (refer to the “Free modules” section in [7]). Secondly, operations such as module lattice addition and intersection can disrupt the pseudo-basis structure. The folklore method addresses this by computing the Hermite Normal Form (HNF) (e.g., see the “Computing arithmetic operations with modules” section in [6]). Alternatively, a more direct approach is to first treat the module as a $\mathbb{Z}$ -lattice, then compute the HNF using a $\mathbb{Z}$ -basis as pseudo-generators to recover a pseudo-basis. Notably, [3] provides a polynomial-time algorithm for computing HNF in the full-rank case. Combining these techniques yields the following lemma. It will be employed in the proofs of [Theorem 2.1](#) and [Theorem 3.3](#). In all quantum reductions in this paper, the “quantum” requirement comes from this lemma, or more precisely, from recovering a free basis from a pseudo-basis. If we permitted the hard problem to be defined using pseudo-bases (a definition which is also meaningful from the perspective of hard problems), this requirement could be removed. However, for the constructive purposes of this paper, we employ the current definition.

Lemma 2.6 (Folklore) *Let $\mathbb{L}$ be a number field, and $B_{\mathcal{O}_{\mathbb{L}}}$ be a $\mathbb{Z}$ basis of $\mathcal{O}_{\mathbb{L}}$. Then there exists a quantum polynomial-time algorithm such that for any full rank free $\mathcal{O}_{\mathbb{L}}$ -module $M \subseteq \mathbb{L}^m$, given a $\mathbb{Z}$ -basis $B_{\mathcal{O}_{\mathbb{L}}}$ of $\mathcal{O}_{\mathbb{L}}$ and a $\mathbb{Z}$ -basis of M , it outputs a free basis of M .*

Lenstra-Silverberg Algorithm. Gentry and Szydlo initially proposed an algorithm in [15] to recover x from x^*x and xR (where R is a certain type of polynomial ring). Later, Lenstra and Silverberg extended this in [18,19,21]. Luo et al. used it to show an algorithm for which, in a certain sense, is a high-dimensional version of the rank 1 module-LIP.

Proposition 2.2 ([24, Proposition 4.1]) *Let $\mathbb{F}$ be a CM-field or a totally real number field with degree n . Let A be the ring of integers of $\mathbb{F}$. [21, Examples 3.7 (i) – (ii)] showed that A is a CM-order. The conjugate automorphism is just the complex conjugation $x \mapsto x^*$, and the trace function is just $\text{Tr}_{\mathbb{F}}$.*

- For $\alpha \in \mathbb{F}$, there is a deterministic polynomial-time algorithm **LS1** that, given A , αA and $\alpha^* \alpha$, then we can find $\alpha \mu(A)$ in polynomial time, where $\mu(A)$ means roots of unity in A .
- For $v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \in \mathbb{F}^2$ and $B \in GL_2(\mathbb{F})$, there is a deterministic polynomial-time algorithm **LS2** that, given A , $B^* B$, $v^* v = v_1 v_1^* + v_2 v_2^*$, and $B^{-1}(A \cdot v)$, then we can find $B^{-1}(\mu(A) \cdot v)$ in polynomial time, where $\mu(A)$ means roots of unity in A .

2.7 Commitment Scheme

Given that the conclusion in [16] provides a good “encapsulation” for using these components to construct commitment schemes, we do not need to provide the full definitions of these schemes. Therefore, we only present the basic commitment definitions to help understanding, while the relevant definitions for Enhanced Linkable Commitment and Dual-Mode Commitment with additional properties are placed in the [Appendix A](#).

Definition 2.6 A commitment scheme Π_{com} consists of three PPT algorithms (**Gen**, **Com**, **Open**) parametrized by a security parameter λ and defined over message space $\mathcal{M}$, commitment space C , and opening space D .

- **Gen**: The algorithm **Gen**(1^λ) outputs the public commitment key Ck .
- **Com**: For any message $m \in \mathcal{M}$, **Com** $_{Ck}(m)$ produces a tuple $(c, d) \in C \times D$, where c is the commitment value and d is the corresponding opening value.
- **Open**: The algorithm **Open** $_{Ck}(c, d)$ returns the message $\tilde{m} \in \mathcal{M} \cup \{\perp\}$, with $\perp$ indicating an invalid commitment.

The correctness of a commitment scheme requires that for any message $m \in \mathcal{M}$, **Open** $_{Ck}(\mathbf{Com}_{Ck}(m)) = m$. The security of a commitment scheme is based on two properties: *hiding* and *binding*, which are defined as follows.

- **Hiding**. Given a commitment value c , it is computationally infeasible for an adversary $\mathcal{A}$ to determine any partial information about the committed message m . Formally, for all PPT adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, there exists a negligible function $\text{negl}(\lambda)$ such that:

$$\Pr \left[b = \tilde{b} \mid \begin{array}{l} Ck \leftarrow \mathbf{Gen}(1^\lambda), (m_0, m_1, \alpha) \leftarrow \mathcal{A}_1(Ck) \\ b \xleftarrow{\$} \{0, 1\}, (c, d) \leftarrow \mathbf{Com}_{Ck}(m_b), \tilde{b} \leftarrow \mathcal{A}_2(c; \alpha) \end{array} \right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

- **Binding**. It is computationally infeasible for an adversary $\mathcal{A}$ to produce a collision, i.e., a triple (c, d, d') where (c, d) and (c, d') are valid commitments for distinct messages $m \neq m'$. Formally, for all PPT adversaries $\mathcal{A}$, there exists a negligible function $\text{negl}(\lambda)$ such that:

$$\Pr \left[\begin{array}{l} m \neq m', \\ m, m' \neq \perp \end{array} \mid \begin{array}{l} Ck \leftarrow \mathbf{Gen}(1^\lambda), (c, d, d') \leftarrow \mathcal{A}(Ck) \\ m \leftarrow \mathbf{Open}_{Ck}(c, d), m' \leftarrow \mathbf{Open}_{Ck}(c, d') \end{array} \right] \leq \text{negl}(\lambda).$$

2.8 Group action

In this subsection, we introduce some definitions related to the group action that are used in [16] to construct the commitment schemes.¹

¹ Our definitions differ slightly from those in [16]. Specifically, we let $\mathbf{X}$ be a single-orbit set and consider a distribution $\mathcal{D}_{\mathbf{X}}$ over it, rather than $\mathcal{D}_{G, \mathbf{X}}$. Our distribution $\mathcal{D}_{\mathbf{X}}$ is induced by first sampling (g, x) from $\mathcal{D}_{G, \mathbf{X}}$ and then generating $g \star x$.

Definition 2.7 (Group Action) Let G be a group with identity element e , and let X be a set. A group action of G on X is an operator $\star : G \times X \rightarrow X$ that satisfies the following conditions for all $g, h \in G$ and $x \in X$:

- The identity element e of G acts trivially on X , i.e., $e \star x = x$.
- The group action is compatible with the group operation, i.e., $g \star (h \star x) = (gh) \star x$.

We denote such a group action by $(G, X, \star)$.

For a group action $(G, X, \star)$, several important concepts are defined as follows:

- The *orbit* of an element $x \in X$, denoted by $\mathcal{O}(x)$, is the set of all elements in X that can be obtained by acting on x with elements of G . Formally, $\mathcal{O}(x) := \{g \star x : g \in G\}$. In this paper, we assume X is a single orbit under the group action for simplicity.
- The *stabilizer* of an element $x \in X$, denoted by $\text{Stab}(x)$, is the subgroup of G consisting of all elements that act trivially on x . Formally, $\text{Stab}(x) := \{g \in G : g \star x = x\}$. In this paper, we only consider the case where the stabilizer is finite and even polynomial of parameter.
- The set of elements in G that map x to another element $y \in X$ is denoted by $\mathcal{I}(x, y)$. Formally, $\mathcal{I}(x, y) := \{g \in G : g \star x = y\}$. It is worth noting that for any $g \in \mathcal{I}(x, y)$, the set $\mathcal{I}(x, y)$ can be expressed as the coset $g \cdot \text{Stab}(x)$.
- We assume a group action $(G, X, \star)$ with an associated efficiently samplable distribution $\mathcal{D}_X$ on X , which we sometimes denote as $(G, X, \star, \mathcal{D}_X)$ for simplicity.

One of the core hard problems in the construction framework is the *search Group Action Stabilizer Problem* (s-GASP), which involves finding a non-trivial element in a stabilizer. Specifically, given an element $x \in X$ such that $\text{Stab}(x) \neq \{e\}$, the objective is to find an element $h \in G$ such that $x = h \star x$ and $h \neq e$. The formal definition of this problem is provided below.

Definition 2.8 (s-GASP) Let $\mathcal{F}$ be a family of group actions such that for a security parameter λ , $\mathcal{F}(1^\lambda)$ returns a group action $(G, X, \star)$ with distribution $\mathcal{D}_X$ over X . The s-GASP assumption requires that for all PPT adversaries $\mathcal{A}$, there is a negligible function $\text{negl}(\lambda)$ such that

$$\Pr[\mathcal{A}(x) \star x = x, \mathcal{A}(x) \neq e \mid x \leftarrow \mathcal{D}_X, \text{Stab}(x) \neq \{e\}] \leq \text{negl}(\lambda).$$

Another important challenging problem is the *decisional Group Action Orbit Problem* (d-GAOP), which requires distinguishing between two classes of orbits. This problem can be viewed as a generalization of the decisional Group Action Inversion Problem (d-GAIP) [13].

Definition 2.9 (d-GAOP) Let $\mathcal{F}$ be a family of group actions such that for a security parameter λ , $\mathcal{F}(1^\lambda)$ returns group actions $(G, X, \star)$ and $(G, X', \star)$ with

distributions $\mathcal{D}_X$ over X and $\mathcal{D}_{X'}$ over X' . The d -GAOP assumption requires that for all PPT adversaries $\mathcal{A}$, there is a negligible function $\text{negl}(\lambda)$ such that

$$\Pr \left[b = \tilde{b} \mid \begin{array}{l} y_0 \leftarrow \mathcal{D}_X, y_1 \leftarrow \mathcal{D}_{X'} \\ b \leftarrow \{0, 1\}, \tilde{b} \leftarrow \mathcal{A}(y_b) \end{array} \right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

Here, we introduce the definition of a **weak re-randomized algorithm** in addition to the original re-randomized algorithm. The only difference between them is that we relax X to $\text{Supp}(\mathcal{D}_X)$. It can be observed that replacing the re-randomized algorithm with the weak re-randomized algorithm in the framework of [16] does not affect the security reduction or the correctness of the scheme, because in practice, the input to the re-randomized algorithm is always an element sampled from $\mathcal{D}_X$. However, the standard re-randomized algorithm provides a worst-case to average-case self-reduction, which cannot be guaranteed by the weak re-randomized algorithm.

As we will see later, the introduction of the weak re-randomized concept is motivated by the fact that, in instantiation, the output size of the re-randomized algorithm is always relatively large, regardless of the input size. By relaxing the requirement, we hope to control the output size for bounded $\text{Supp}(\mathcal{D}_X)$.

Definition 2.10 (Re-Randomized Algorithm) For a group action $(G, X, \star, \mathcal{D}_X)$, a (resp. weak) re-randomized algorithm R takes as input $x \in X$ and outputs a pair $(g \star x, g) \in X \times G$ according to a distribution, denoted as $R(x)$, such that for any $x, x' \in X$ (resp. $\text{Supp}(\mathcal{D}_X)$), and $(x'', g) \leftarrow R(x)$

- the marginal distributions of the first variable are identical for $R(x)$ and $R(x')$,
- g is uniformly distributed on $\mathcal{I}(x, x'')$.

We note that there exists a sufficient condition for a better use of the re-randomized algorithm. Its proof is provided in the [Appendix C](#).

Lemma 2.7 For a group action $(G, X, \star, \mathcal{D}_X)$, an algorithm R is a (resp. weak) re-randomized algorithm if it takes as input $x \in X$ and outputs a pair $(g \star x, g) \in X \times G$ according to a distribution, denoted as $R(x)$, such that:

- For any $x, x' \in X$ (resp. $\text{Supp}(\mathcal{D}_X)$), $h \in \mathcal{I}(x, x')$, the distributions of $R(h \star x)[2]$ and $R(x)[2] \cdot h^{-1}$ are same.

Another key component is the Extractor E , whose function is to extract the hard core from the isomorphisms $\text{Iso}(x, x'')$ between two elements. Combined with the aforementioned Re-Randomized Algorithm, we obtain a paradigm for constructing commitment schemes as in [Protocol 2.1](#).

The different properties possessed by the Extractor will help us construct commitment schemes with varying functionalities. For example, when it can uniformly extract information from $\text{Iso}(x, x'')$, the corresponding commitment scheme achieves perfect hiding because the re-randomized algorithm guarantees the perfect randomness of $c_2 = y'$ and the randomness of g on $\text{Iso}(x, x'')$, and

the uniformly extracting property further ensures the perfect randomness of c_1 . Breaking the binding property, on the other hand, means finding two distinct d_1, d_2 such that $d_i \star y = y'$, which in turn implies finding a non-trivial element in $\text{Stab}(y)$. When it also has additional homomorphic properties, it can endow the commitment with certain homomorphic features, and so on.

Commitment Scheme 2.1 : $\Pi_{\text{com}} = (\text{Gen}, \text{Com}, \text{Open})$ for a security parameter λ and message space M .

- **Gen**(1^λ): Sample $y \leftarrow \mathcal{D}_X$ (resp. $\mathcal{D}_{X'}$) and output $Ck = y$.
- **Com** $_y(m)$: For a message $m \in M$, sample $(y', g) \leftarrow \mathcal{R}(y)$ such that $g \star y = y'$. Compute $c = (c_1, c_2) = (E(g) \cdot m, y')$ and set $d = g$. Output (c, d) .
- **Open** $_y(c, d)$: If $d \star y = c_2$, output $\tilde{m} = E(d)^{-1} \cdot c_1$; otherwise, output $\perp$.

Definition 2.11 (Extractor) For group actions $(G, X, \star)$, $(G, X', \star)$ and M is an abelian group. An extractor is defined as a deterministic and efficient algorithm $E : G \rightarrow M$, where $|M| \geq 2$. It may optionally possess the following properties:

- *Uniformly extracting property (on X):* for any $y', y'' \in X$, it holds that $E(g)$ is uniformly distributed on M for $g \leftarrow_{\$} \mathcal{I}(y', y'')$.
- *Homomorphic property:* $E : G \rightarrow M$ is a surjective group homomorphism, i.e., for any $g_0, g_1 \in G$, $E(g_0) \cdot E(g_1)^{-1} = E(g_0 \cdot g_1^{-1})$.
- *Local constant property (on X'):* for any $y', y'' \in X'$, it holds that $E(g)$ is a constant for $g \in \mathcal{I}(y', y'')$.
- *Trapdoor property (on X):* exist a PPT algorithm F that takes $(u, g, x, \text{Stab}(X)) \in M \times G \times X \times \mathcal{P}(G)$ as input and outputs $g' \in G$, such that: for any $u \in M$, $y', y'' \in X$, $g \in \mathcal{I}(y', y'')$, the output g' of $F(u, g, y', \text{Stab}(y'))$ satisfies $g' \in \mathcal{I}(y', y'')$ and $E(g') = u$; the distributions of $(E(g), g)$ and (u, g') are identical for any $y', y'' \in X$, where $g \leftarrow_{\$} \mathcal{I}(y', y'')$, $u \leftarrow_{\$} M$, $g' \leftarrow F(u, g, y', \text{Stab}(y'))$.

Indeed, the uniformly extracting property and homomorphic property imply the Trapdoor property.

Lemma 2.8 For the group action $(G, X, \star)$, if an extractor E satisfies the uniformly extracting property (on X) and the homomorphic property, then it has the Trapdoor property.

Proof. It suffices to construct F as follows: on input $(u, g, y', \text{Stab}(y'))$, sample uniformly at random an element h from the set $\{h \in \text{Stab}(y') \mid E(h) = E(g)^{-1}u\}$, and output $g' = gh$. This satisfies $E(g') = u$.

When u is uniformly distributed over M , $E(g)^{-1}u$ is also uniformly distributed over M . Moreover, since $E(U(\mathcal{I}(y', y'')))$ is uniform over M , the sampled g' is uniformly distributed over $\mathcal{I}(y', y'')$. Consequently, the distributions of $(E(g), g)$ and (u, g') are identical. $\square$

Theorem 2.2 ([16, Theorem 3.1,3.2,4.1,4.2]) *Suppose that group action $(G, X, \star, \mathcal{D}_X)$ satisfies the s -GASP assumption, $(G, X, \star, \mathcal{D}_X)$ and $(G, X', \star, \mathcal{D}_{X'})$ satisfies the d -GAOP assumption, R is a weak re-randomized algorithm, and E is an extractor. Then*

- *If E has the uniformly extracting property on X , then Protocol 2.1 for X is a perfectly hiding and computationally binding commitment scheme based on the s -GASP assumption over $(G, X, \star, \mathcal{D}_X)$. We refer to this as the basic scheme.*
- *If E has the uniformly extracting property on X and local constant property on X' , then Protocol 2.1 for X' is a computationally hiding and perfect binding commitment scheme based on the s -GASP assumption over $(G, X, \star, \mathcal{D}_X)$. We refer to this as the dual scheme.*
- *If E has the uniformly extracting property and homomorphic property on X , then Protocol A.1 is an enhanced linkable commitment.*
- *Assume that for some representative element x in X , its stabilizer is known, and the distribution $\mathcal{D}_X$ is generated by sampling an element g from G and outputting $g \star x$. If E has the uniformly extracting property and trapdoor property on X and local constant property on X' , then Protocol A.2 is a dual mode commitment.*

3 Module-LIP and module-LAP on certain modules

In this section and the next, unless otherwise specified, we follow the setup in HAWK by setting $\mathbb{L} = \mathbb{Q}[X]/(\Phi_{2^d}(X)) = \mathbb{Q}[X]/(X^{2^{d-1}} + 1)$, with $n = \deg(\mathbb{L})$.

3.1 Aut of $\mathbf{Q}_k^{(0)}$ and $\mathbf{Q}_k^{(2)}$

Due to space constraints, the proofs in this subsection are provided in the Appendix C.

Lemma 3.1 *For $k > 0$, $\text{Aut}_{\mathcal{O}_L}(\mathbf{Q}_k^{(0)}) = \left\{ \begin{pmatrix} \xi_1 & 0 \\ 0 & \xi_2 \end{pmatrix} \mid \xi_1, \xi_2 \in \mu(\mathbb{L}) \right\}$.*

Lemma 3.2 *For $k > 2$, $M = \begin{pmatrix} 1 & 1 + X^2 \\ 0 & -1 \end{pmatrix}$,*

$$\text{Aut}_{\mathcal{O}_L}(\mathbf{Q}_k^{(2)}) = M^{-1} \left\{ \begin{pmatrix} \xi_1 & 0 \\ \xi_2 - \xi_1 & \xi_2 \end{pmatrix} \mid \xi_1, \xi_2 \in \mu(\mathbb{L}), \xi_1/\xi_2 = X^{2k}, k \in \mathbb{Z} \right\} M.$$

3.2 LIP to LAP reduction on $[\mathbf{Q}_k^{(0)}]$

Theorem 3.1 *There is a reduction from module-LIP $_{[\mathbf{Q}_k^{(0)}]}$ to module-LAP $_{[\mathbf{Q}_k^{(0)}]}$. In particular, our reduction preserves the instance of the problems i.e. for any $\mathbf{Q} \in [\mathbf{Q}_k^{(0)}]$ we can reduce module-LIP $_{\mathbf{Q}, \mathbf{Q}_k^{(0)}}$ to module-LAP $_{\mathbf{Q}}$.*

For $\mathbf{Q} = U^* \mathbf{Q}_k^{(0)} U$, any module automorphism can be written as $U^{-1} P U$ for some $P \in \text{Aut}_{\mathcal{O}_L}(\mathbf{Q}_k^{(0)})$. By Lemma 2.3 and Lemma 3.1, the possible candidates for P are known and are polynomial. Therefore, by enumeration, we can assume that given the input module lattice automorphism $U^{-1} P U$, we have guessed P .

We aim to replace $(J_2, (BU)^{-1} J_2 (BU))$ with $(P, U^{-1} P U)$ in order to apply the techniques from [24]. A sketch of this step can be found in Algorithm 1. There are some subtle adjustments here, due to the fact that the automorphisms we defined and the pseudo-automorphisms considered in [24] cannot be directly converted between each other.

Algorithm 1: FindU(Informal)

Require: $P \in \text{Aut}_{\mathcal{O}_L}(\mathbf{Q}_k^{(0)}) \setminus \mu(\mathbb{L}) I_2$; $U^{-1} P U$ and $\mathbf{Q} = U^* \mathbf{Q}_k^{(0)} U$ for which $U \in \text{GL}_2(\mathcal{O}_L)$

Ensure: A congruence matrix U between $\mathbf{Q}_k^{(0)}$ and $\mathbf{Q}$

- 1: $\lambda \leftarrow$ one of eigenvalues of P
 - 2: $M \leftarrow \ker(P - \lambda I_2) \cap \mathcal{O}_L^2$, $U^{-1} M \leftarrow \ker(U^{-1} P U - \lambda I_2) \cap \mathcal{O}_L^2$
 - 3: Write $B_k^{(0)} M$ as $\mathcal{I} \cdot v$
 - 4: $(B_k^{(0)} U)^{-1} (\mathcal{O}_L \cdot v) \leftarrow \mathcal{I}^{-1} \cdot U^{-1} M$
 - 5: $\mu(\mathbb{L}) \cdot (B_k^{(0)} U)^{-1} v \leftarrow \text{LS2}(\mathcal{O}_L, v^* v, U^* \mathbf{Q}_k^{(0)} U, (B_k^{(0)} U)^{-1} (\mathcal{O}_L \cdot v))$
 - 6: $B_k^{(0)} U \leftarrow (v | t_*(v)) ((B_k^{(0)} U)^{-1} v | (B_k^{(0)} U)^{-1} t_*(v))^{-1}$
 - 7: **return** $(B_k^{(0)})^{-1} (B_k^{(0)} U)$.
-

By Lemma 2.3 and Lemma 3.1, the structure of P can be classified into two types: diagonal and anti-diagonal. When P is a diagonal matrix, this operation can be easily accomplished. The process is described in the following proposition.

Proposition 3.1 Assume $U \in \text{GL}_2(\mathcal{O}_L)$. Given $\mathbf{Q} = U^* \mathbf{Q}_k^{(0)} U$, $\xi_1 \neq \xi_2 \in \mu(\mathbb{L})$, and $P_U := U^{-1} P U$ where $P := \begin{pmatrix} \xi_1 & 0 \\ 0 & \xi_2 \end{pmatrix}$, we can find U in polynomial time of the size of input.

Proof. We first compute $\ker(P_U - \xi_1 I_2) \cap \mathcal{O}_L^2$ by Lemma 2.5 in polynomial time, and it's just

$$\begin{aligned} \ker(U^{-1} \begin{pmatrix} 0 & 0 \\ 0 & \xi_2 - \xi_1 \end{pmatrix} U) \cap \mathcal{O}_L^2 &= U^{-1} (e_1 \mathbb{L}) \cap U^{-1} \mathcal{O}_L^2 = U^{-1} (e_1 \mathbb{L} \cap \mathcal{O}_L^2) \\ &= U^{-1} e_1 \mathcal{O}_L = (B_k^{(0)} U)^{-1} (B_k^{(0)} e_1 \mathcal{O}_L). \end{aligned}$$

where $e_1 = (1, 0)^T$. Also we can compute $\tau_{(B_k^{(0)} U)} = (B_k^{(0)} U)^{-1} J_2 t_*(B_k^{(0)} U)$ by Proposition 2.1 using $\mathbf{Q}$.

Then using LS2 (Proposition 2.2) for input $\mathcal{O}_L$, $(B_k^{(0)} e_1)^* (B_k^{(0)} e_1)$, $(B_k^{(0)} U)^{-1} (B_k^{(0)} e_1 \mathcal{O}_L)$, $(B_k^{(0)} U)^* (B_k^{(0)} U) = \mathbf{Q}$, we can find $\mu(\mathbb{L}) \cdot (B_k^{(0)} U)^{-1} (B_k^{(0)} e_1)$.

Next, for every $w \in \mu(\mathbb{L}) \cdot (B_k^{(0)}U)^{-1}(B_k^{(0)}e_1)$, we have

$$(B_k^{(0)}e_1 | J_2 t_* B_k^{(0)}e_1)(w | \tau_{(B_k^{(0)}U)} w)^{-1}.$$

Then we can find the $B_k^{(0)}U$ ((when $w = e_1$)) and then compute U . $\square$

However, when $k = 0$, P might be an anti-diagonal matrix, and consequently, there might be no eigenvalues of P in $\mathbb{L}$. In order to use the eigenspaces to decompose the $\mathcal{O}_{\mathbb{L}}^2$, it is necessary to extend the field L by adjoining the eigenvalues of P which is the square root of a root of unity in L . In this case, we must also prove that the extended field remains a CM field and that an integral basis of its ring of integers can still be efficiently computed. When $\mathbb{L}$ is a cyclotomic field, this follows easily from the structure of cyclotomic fields since the extended field is also a cyclotomic field with bounded degree.

Proposition 3.2 *Assume $U \in GL_2(\mathcal{O}_{\mathbb{L}})$. Given U^*U , $\xi_1, \xi_2 \in \mu(\mathbb{L})$, and $P_U := U^{-1}PU$ where $P := \begin{pmatrix} 0 & \xi_1 \\ \xi_2 & 0 \end{pmatrix}$, we can find U in polynomial time of the size of input.*

Proof. Let $\xi := \xi_1 \cdot \xi_2$, and

$$\mathbb{F} := \begin{cases} \mathbb{L}[Y]/(Y^2 - \xi) & \text{if } Y^2 - \xi \text{ doesn't have roots in } \mathbb{L}; \\ \mathbb{L} & \text{otherwise, and take } Y \text{ s.t. } Y^2 = \xi. \end{cases}$$

For the former, we know that $\{Y^i \mid i = 0, \dots, 2n-1\}$ is a $\mathbb{Z}$ -basis of $\mathcal{O}_{\mathbb{F}}$. Thus we can compute $\ker(P_U - Y) \cap \mathcal{O}_{\mathbb{F}}^2$ by [Lemma 2.5](#) in polynomial time, and it's just

$$\ker(U^{-1} \begin{pmatrix} -Y & \xi_1 \\ \xi_2 & -Y \end{pmatrix} U) \cap \mathcal{O}_{\mathbb{F}}^2 = U^{-1}(v\mathbb{F}) \cap U^{-1}\mathcal{O}_{\mathbb{F}}^2 = U^{-1}(v\mathbb{F} \cap \mathcal{O}_{\mathbb{F}}^2) = U^{-1}v\mathcal{O}_{\mathbb{F}}.$$

where $v = (\xi_1, Y)^T$. Also we can compute $U^{-1}J_2 t_* U$ by [Proposition 2.1](#) using U^*U . Then using [LS2](#) for input $\mathcal{O}_{\mathbb{L}}$, v^*v , $U^{-1}v\mathcal{O}_{\mathbb{L}}$ and U^*U , we can find $\mu(\mathbb{L}) \cdot U^{-1}v$. Next, for every $w \in \mu(\mathbb{L}) \cdot U^{-1}v$, we compute $(v | J_2 t_* v)(w | U^{-1}J_2 t_* U w)^{-1}$. Then we can find the U (when $w = v$). $\square$

Proof (of [Theorem 3.1](#)). Given $\mathbf{Q} = U^*\mathbf{Q}_k^{(0)}U$ for some $U \in GL_2(\mathcal{O}_{\mathbb{L}})$, and a non-trivial automorphism $P_U \in \text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q})$, we know P_U has the form $U^{-1}PU$ for some $P \in \text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q}_k^{(0)}) \setminus \mu(\mathbb{L})I_2$.

Iterate over $\tilde{P} \in \text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q}_k^{(0)}) \setminus \mu(\mathbb{L})I_2$. By [Lemma 3.1](#), when $k > 0$,

$$\text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q}_k^{(0)}) = \left\{ \begin{pmatrix} \xi_1 & 0 \\ 0 & \xi_2 \end{pmatrix} \mid \xi_1, \xi_2 \in \mu(\mathbb{L}) \right\},$$

and when $k = 0$, $\text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q}_k^{(0)}) = \mathcal{U}_2(\mathcal{O}_{\mathbb{L}})$. If $\tilde{P}$ is a diagonal matrix, input $\mathbf{Q}$, P_U , and $\tilde{P}$ into [Proposition 3.1](#); if $\tilde{P}$ is an anti-diagonal matrix (implies $k = 0$),

input them into [Proposition 3.2](#), and verify whether the inverse of the output matrix is an isomorphism from $\mathbf{Q}$ to $\mathbf{Q}_k^{(0)}$.

Correctness: When $P = \tilde{P}$, by [Proposition 3.1](#), [Proposition 3.2](#), it will output the right U . U^{-1} is an isomorphism from $\mathbf{Q}$ to $\mathbf{Q}_k^{(0)}$.

Complexity: The number of iterations is less than the cardinality of $\text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q}_k^{(0)})$, which is bounded by $O(n^4)$. Each loop runs in polynomial time by [Proposition 3.1](#) and [Proposition 3.2](#).

Therefore, the overall time complexity is polynomial. $\square$

When $\mathbb{L}$ is a general CM field, we also have a reduction from $\mathcal{O}_{\mathbb{L}}^2$ -LIP to $\mathcal{O}_{\mathbb{L}}^2$ -LAP. This is not immediately obvious, precisely due to the reason mentioned earlier: when P is an anti-diagonal matrix, we need to consider the structure of the extended field. Due to space constraints, we have placed the proofs in the [Appendix B](#).

Theorem 3.2 *Let $\mathbb{L}$ be a CM number field. Given the parameter $\mathbf{L}$, there is a reduction from $\mathcal{O}_{\mathbb{L}}^2$ -LIP to $\mathcal{O}_{\mathbb{L}}^2$ -LAP. In particular, our reduction preserves the instance of the problems i.e. for any $\mathbf{Q} \in [I_2]$ we can reduce module-LIP $_{\mathbf{Q}, I_2}$ to module-LAP $_{\mathbf{Q}}$.*

3.3 Hardness of module-LIP $_{[\mathbf{Q}_k^{(0)}]}$ for $k = O(\log n)$

In the previous subsection, we proved that $\text{module-LAP}_{[\mathbf{Q}_k^{(0)}]} \geq \text{module-LIP}_{[\mathbf{Q}_k^{(0)}]}$. In this subsection, we further prove that $\text{module-LIP}_{[\mathbf{Q}_k^{(0)}]} = \text{module-LIP}_{[I_2]}$, i.e., $\mathcal{O}_{\mathbb{L}}^2$ -LIP, for $k = O(\log n)$.

Theorem 3.3 *For $k = O(\log n)$, there is a (quantum) reduction from $\mathcal{O}_{\mathbb{L}}^2$ -LIP to module-LIP $_{[\mathbf{Q}_k^{(0)}]}$.*

Due to space constraints, we have placed the proofs in the [Appendix C](#).

Remark 4. The use of quantum computation arises in the step when showing that $\text{module-LIP}_{[\mathbf{Q}_k^{(0)}]} \geq \text{module-LIP}_{[\mathbf{Q}_\ell^{(0)}]}$, fundamentally because we cannot compute $\mathcal{O}_{\mathbb{L}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \mathcal{O}_{\mathbb{L}} \begin{pmatrix} 0 \\ \Pi^{k+1} \end{pmatrix}$ from $\mathcal{O}_{\mathbb{L}} \begin{pmatrix} \Pi \\ 0 \end{pmatrix} + \mathcal{O}_{\mathbb{L}} \begin{pmatrix} 0 \\ \Pi^{k+1} \end{pmatrix}$ while preserving the free basis representation. This requires invoking [Lemma 2.6](#). Therefore, if we only require the reduction for the geometric version of module-LIP, the quantum step can be omitted.

Remark 5. Note that for any $\text{poly}(n) \geq \ell \geq k$, the reduction $\text{module-LIP}_{[\mathbf{Q}_k^{(0)}]} \geq \text{module-LIP}_{[\mathbf{Q}_\ell^{(0)}]}$ implies that $\mathcal{O}_{\mathbb{L}}^2$ -LIP $\geq \text{module-LIP}_{[\mathbf{Q}_k^{(0)}]}$. Combined with the above theorem, we have $\text{module-LIP}_{[\mathbf{Q}_k^{(0)}]} = \mathcal{O}_{\mathbb{L}}^2$ -LIP, for $k = O(\log n)$.

3.4 Hardness of module- $\Delta\text{LIP}_{\mathbb{L}, [\mathbf{Q}_{2k}^{(0)}], [\mathbf{Q}_k^{(2)}]}$

Lemma 3.3 For $\mathbb{L} = \mathbb{Q}[X]/(X^{2^{d-1}} + 1)$, let $\mathbb{K} = \mathbb{Q}(X^2) \subseteq \mathbb{L}$. Then module- $\Delta\text{LIP}_{\mathbb{L}, [\mathbf{Q}_{2k}^{(0)}], [\mathbf{Q}_k^{(2)}]}$ is at least hard as $\Delta_{k, \mathbb{K}}$.

Proof. It suffices to prove that the matrices $[\mathbf{Q}_k^{(0)}] = \left[\begin{pmatrix} 1 & 0 \\ 0 & (1+X^2)^k \end{pmatrix}^* \begin{pmatrix} 1 & 0 \\ 0 & (1+X^2)^k \end{pmatrix} \right]$, and $[\mathbf{Q}_k^{(1)}] = \left[\begin{pmatrix} 1 & 0 \\ (1+X^2)^{k-1} & (1+X^2)^k \end{pmatrix}^* \begin{pmatrix} 1 & 0 \\ (1+X^2)^{k-1} & (1+X^2)^k \end{pmatrix} \right]$, defined over $\mathbb{K}$ are contained within $[\mathbf{Q}_{2k}^{(0)}]_{\text{gl}} = \left[\begin{pmatrix} 1 & 0 \\ 0 & (1+X)^{2k} \end{pmatrix}^* \begin{pmatrix} 1 & 0 \\ 0 & (1+X)^{2k} \end{pmatrix} \right]_{\text{gl}}$ and $[\mathbf{Q}_k^{(2)}]_{\text{gl}} = \left[\begin{pmatrix} 1 & 0 \\ (1+X^2)^{k-1} & (1+X^2)^k \end{pmatrix}^* \begin{pmatrix} 1 & 0 \\ (1+X^2)^{k-1} & (1+X^2)^k \end{pmatrix} \right]_{\text{gl}}$ defined over $\mathbb{L}$, respectively. Note that $\text{SL}_2(\mathcal{O}_{\mathbb{K}}) \subseteq \text{GL}_2(\mathcal{O}_{\mathbb{L}})$, so we only need to show that there exists $U_0 \in \text{GL}_2(\mathcal{O}_{\mathbb{L}})$ such that

$$\begin{pmatrix} 1 & 0 \\ 0 & (1+X)^{2k} \end{pmatrix} U_0 = \begin{pmatrix} 1 & 0 \\ 0 & (1+X^2)^k \end{pmatrix},$$

i.e.,

$$\begin{pmatrix} 1 & 0 \\ 0 & (1+X)^{2k} \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ 0 & (1+X^2)^k \end{pmatrix} \in \text{GL}_2(\mathcal{O}_{\mathbb{L}}),$$

and there exists $U_1 \in \text{GL}_2(\mathcal{O}_{\mathbb{L}})$ such that

$$\begin{pmatrix} 1 & 0 \\ (1+X^2)^{k-1} & (1+X^2)^k \end{pmatrix} U_1 = \begin{pmatrix} 1 & 0 \\ (1+X^2)^{k-1} & (1+X^2)^k \end{pmatrix}.$$

The latter is obvious. The former follows from $(1+X^2)(1+X)^{-2} \in \mathcal{O}_{\mathbb{L}}^\times$. $\square$

Theorem 2.1 reduces $\mathcal{O}_{\mathbb{K}}^2\text{-LIP}$ to $\{\Delta_{k, \mathbb{K}}, k = 1, \dots, n^2/2\}$. Combined with the lemma above, we can reduce $\mathcal{O}_{\mathbb{K}}^2\text{-LIP}$ to $\{\text{module-}\Delta\text{LIP}_{\mathbb{L}, [\mathbf{Q}_{2k}^{(0)}], [\mathbf{Q}_k^{(2)}]}, k = 1, \dots, n^2/2\}$.

For any cyclotomic field $\mathbb{F}$ of power-of-two degree, $\mathcal{O}_{\mathbb{F}}^2\text{-LIP}$ can be viewed as a problem parameterized by $\deg(\mathbb{F})$. From a practical solving perspective, for any $k > 1$, $\{\mathbf{Q}_{2k}^{(0)}, \mathbf{Q}_k^{(2)}\}$ lie in the same genus and even in the same special genus, and there are currently no known algorithms for solving them other than via SVP. From a hardness perspective, if the problem parameterized by n is considered hard, then the problem parameterized by $n/2$ can still be regarded as hard. Therefore, under the assumption that $\mathcal{O}_{\mathbb{F}}^2\text{-LIP}$ is hard, $\{\text{module-}\Delta\text{LIP}_{\mathbb{L}, [\mathbf{Q}_{2k}^{(0)}], [\mathbf{Q}_k^{(2)}]}, k = 1, \dots, n^2/2\}$ provides potential hard instances of module- ΔLIP . Although we lack theoretical results to definitively establish which problems achieve the hardness of $\mathcal{O}_{\mathbb{F}}^2\text{-LIP}$, based on the reduction for the search version, we heuristically argue that problems corresponding to smaller values of k are more representative of the problem's inherent hardness.

4 Commitment Schemes based on module-LIP

4.1 Choice of instances

Group Action Based on Module Lattice. A naive idea for constructing a structured version in [16] is to replace $GL_n(\mathbb{Z})$ with $GL_2(\mathcal{O}_{\mathbb{L}})$. However, when constructing the re-randomized algorithm, obstacles arise because, unlike general lattices, which can use efficient algorithm (like LLL) to find an integral basis, there is currently no efficient algorithm on classical computers for finding a $\mathcal{O}_{\mathbb{L}}$ -basis for module lattices. This issue is also reflected in the self-reduction of the module lattice isomorphism problem. To bypass this difficulty, [10] considers $SL_2(\mathcal{O}_{\mathbb{L}})$. If we adopt their strategy, we face another obstacle when constructing deterministic extractors (since we cannot still use $\det$ as an extractor as in [16]). To meet both requirements simultaneously, we consider such a subgroup.

Definition 4.1 *We define the Generalized Special Linear group over $\mathcal{O}_{\mathbb{L}}$ as $GSL_2(\mathcal{O}_{\mathbb{L}}) := \{M \in GL_2(\mathcal{O}_{\mathbb{L}}) | \det(M) \in \mu(\mathbb{L})\}$ and the projective group of it is $PGSL_2(\mathcal{O}_{\mathbb{L}}) := GSL_2(\mathcal{O}_{\mathbb{L}})/\mu(\mathbb{L})I_2$.*

Let $\mathbf{Q} \in \mathcal{H}_n^{>0}(\mathcal{O}_{\mathbb{L}})$ be a positive definite Hermitian form over $\mathcal{O}_{\mathbb{L}}$, and let $[\mathbf{Q}]$ denote the set of Hermitian forms equivalent to $\mathbf{Q}$, i.e., $[\mathbf{Q}]_{\text{sl}} := \{\mathbf{V}^* \mathbf{Q} \mathbf{V} : \mathbf{V} \in SL_n(\mathcal{O}_{\mathbb{L}})\}$, $[\mathbf{Q}]_{\text{gsl}} := \{\mathbf{V}^* \mathbf{Q} \mathbf{V} : \mathbf{V} \in GSL_n(\mathcal{O}_{\mathbb{L}})\}$. We can define the group action $(PGSL_n(\mathcal{O}_{\mathbb{L}}), [\mathbf{Q}]_{\text{gsl}}, \star)$ as ²:

$$\mathbf{Q}' \star \mathbf{V} = \mathbf{V}^* \mathbf{Q}' \mathbf{V} \quad \text{for any } \mathbf{V} \in PGSL_n(\mathcal{O}_{\mathbb{L}}), \mathbf{Q}' \in [\mathbf{Q}]_{\text{gsl}}. \quad (1)$$

This group action is closely related to free module lattice isomorphisms and automorphisms. Note that any automorphism of a rank-2 Hermitian form over $\mathcal{O}_{\mathbb{L}}$ is contained in $GSL_2(\mathcal{O}_{\mathbb{L}})$. The purpose of considering the projective group is to exclude trivial automorphisms of the form $\mu(\mathbb{L})I_n$ from the stabilizer of the group action. Consequently, the s-GASP for this group action on $[\mathbf{Q}]_{\text{gsl}}$ is equivalent to module-LAP. Similarly, the d-GAOP for $[\mathbf{Q}_1]_{\text{gsl}}$ and $[\mathbf{Q}_2]_{\text{gsl}}$ is equivalent to module- Δ LIP $_{\mathcal{D}_{[\mathbf{Q}_1]}, \mathcal{D}_{[\mathbf{Q}_2]}}$, where the conversion between isomorphism classes can be handled analogously to Remark 1.

Here, we focus on the case where $n = 2$ and $\mathbf{X} = [\mathbf{Q}_{\mathbf{k}}^{(0)}]_{\text{gsl}}, \mathbf{X}' = [\mathbf{Q}_{\mathbf{k}}^{(2)}]_{\text{gsl}}$. And assume the distribution $\mathcal{D}_{\mathbf{X}}$ (resp. $\mathcal{D}_{\mathbf{X}'}$) is generated by sampling an element g from $PGSL_2(\mathcal{O}_{\mathbb{L}})$ and outputting $g \star \mathbf{Q}_{\mathbf{k}}^{(0)}$ (resp. $\mathbf{Q}_{\mathbf{k}}^{(2)}$).

4.2 Re-randomized algorithm for $(PGSL_2(\mathcal{O}_{\mathbb{L}}), [\mathbf{Q}_{\mathbf{k}}^{(0)}]_{\text{gsl}}, \star)$

As mentioned earlier, the main difficulty in structuring the instantiation in [16] lies in constructing a suitable re-randomized algorithm. We make adaptations to the construction in [10] for self-reduction to obtain the required re-randomize

² The choice of representative in $GSL_n(\mathcal{O}_{\mathbb{L}})/\mu(\mathbb{L})I_n$ does not matter for this group action since $\mathbf{V}^* \mathbf{Q} \mathbf{V} = (\xi \mathbf{V})^* \mathbf{Q} (\xi \mathbf{V})$ for any $\xi \in \mu(\mathbb{L})$.

algorithm for $(\text{PGSL}_2(\mathcal{O}_{\mathbb{L}}), [\mathbf{Q}_k^{(0)}]_{\text{gs1}, \star})$. Note that $\mathbf{Q}_k^{(0)}$ is a diagonal matrix, for which we have $[\mathbf{Q}_k^{(0)}]_{\text{gs1}} = [[\mathbf{Q}_k^{(0)}]_{\text{sl}}] \triangleq [\mathbf{Q}_k^{(0)}]$.

We rewrite [10, Section 6, Lemmas 4,5] to better highlight the algorithmic properties involved, and provide the proofs in the [Appendix C](#).

Lemma 4.1 (Adapted [10, Section 6 Lemma 4]) *Assume $\mathbf{Q}_0 \in \mathcal{H}_2(\mathbb{L})$ is diagonal and $S \subseteq [\mathbf{Q}_0]$, and $T = \sup\{\|\widetilde{B_{\text{rot}(\mathbf{Q})}}\|/\pi \cdot \sqrt{\log(2n+4)} : \mathbf{Q} \in S\}$ (T could be ∞). Under [Heuristic 1](#) for $\mathbf{Q}_0$, for $\sigma \geq \min\{2^{\Theta(n)}\lambda_{2n}(\mathbf{Q}_0), T\}$, there exists an efficient randomized algorithm R_0^S that takes any $\mathbf{Q} \in S$ as input and outputs $(\mathbf{R}, \mathbf{U})$ such that $(\mathbf{R} = \mathbf{U}^* \mathbf{Q} \mathbf{U}, \mathbf{U}) \in [\mathbf{Q}]_{\text{sl}} \times \text{SL}_2(\mathcal{O}_{\mathbb{L}})$. We omit the superscript S whenever it does not cause ambiguity. The distribution of $R_0(\mathbf{Q})[2]$ is given by:*

– For any $\mathbf{U} = (u_1|u_2) \in \text{SL}_2(\mathcal{O}_{\mathbb{L}})$,

$$\Pr[R_0(\mathbf{Q})[2] = \mathbf{U}] = \begin{cases} \rho_{\mathbf{Q}, \sigma}(u_1)/\rho_{\mathbf{Q}_0, \sigma}((\mathcal{O}_{\mathbb{L}}^2)^*), & \text{if } u_2 \text{ is reduced with respect} \\ & \text{to } u_1 \text{ and } \mathbf{Q}; \\ 0, & \text{otherwise,} \end{cases}$$

with the following property:

– For any $\mathbf{V} \in \text{SL}_2(\mathcal{O}_{\mathbb{L}})$ s.t. $\mathbf{V}^* \mathbf{Q} \mathbf{V} \in S$, the distribution of $\mathbf{V}^{-1} \cdot R_0(\mathbf{Q})[2]$ is the same as that of $R_0(\mathbf{V}^* \mathbf{Q} \mathbf{V})[2]$.

Here $\text{Reduce}(\mathbf{Q}, x, y) := y - \left[\frac{x^* \cdot \mathbf{Q} \cdot y}{x^* \cdot \mathbf{Q} \cdot x} \right] \cdot x$ and we say y is reduced with respect to x and $\mathbf{Q}$ if $\left[\frac{x^* \cdot \mathbf{Q} \cdot y}{x^* \cdot \mathbf{Q} \cdot x} \right] = 0$ in which $[\cdot] : a_0 + a_1X + \dots + a_{n-1}X^{n-1} \mapsto \lceil a_0 \rceil + \lceil a_1 \rceil X + \dots + \lceil a_{n-1} \rceil X^{n-1}$.

Algorithm 2: R_0^S (adapted from [10, Algorithm 6]):

Require: Conductor $m = 2^\kappa$ cyclotomic $\mathbb{L}$, $\mathbf{Q} \in S$, $\sigma \geq \min\{2^{\Theta(n)}\lambda_{2n}(\mathbf{Q}_0), T\}$

Ensure: $\mathbf{R} \in [\mathbf{Q}]_{\text{sl}}$ and $\mathbf{U} \in \text{SL}_2(\mathcal{O}_{\mathbb{L}})$ such that $\mathbf{R} = \mathbf{U}^* \cdot \mathbf{Q} \cdot \mathbf{U}$

```

1: Parse  $u_1 \leftarrow D_{\mathbf{Q}, \sigma}$  as  $u_1 = (f, g)^T \in \mathcal{O}_{\mathbb{L}}^2$ 
2: if HermiteSolve( $\mathbb{L}, u_1$ ) returns  $\perp$  then
3:   Restart
4: else
5:    $\widetilde{u}_2 \leftarrow \text{HermiteSolve}(\mathbb{L}, u_1)$ 
6: end if
7: Let  $u_2 = (F_{\mathbf{Q}}, G_{\mathbf{Q}})^T \leftarrow \text{Reduce}(\mathbf{Q}, u_1, \widetilde{u}_2)$ 
8: Let  $\mathbf{U} = (u_1 | u_2)$  and  $\mathbf{R} = \mathbf{U}^* \cdot \mathbf{Q} \cdot \mathbf{U}$ 
9: return  $(\mathbf{R}, \mathbf{U})$ 
```

There are some observations regarding the randomized algorithm R_0 when considered on a diagonal Hermitian form.

Lemma 4.2 Assume $\mathbf{Q}_0 \in \mathcal{H}_2(\mathbb{L})$ is diagonal. For any $\xi \in \mu(\mathbb{L}) = \{1, X, \dots, X^{n-1}\}$, let $\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 0 & \xi \end{pmatrix}$ and $\mathbf{Y} = \begin{pmatrix} \xi & 0 \\ 0 & \xi^{-1} \end{pmatrix}$. Then for any $\mathbf{U} \in \text{SL}_2(\mathcal{O}_{\mathbb{L}})$, we have

$$\Pr[\mathbf{R}_0(\mathbf{Q}_0)[2] = \mathbf{X}^{-1}\mathbf{U}\mathbf{X}] = \Pr[\mathbf{R}_0(\mathbf{Q}_0)[2] = \mathbf{U}] = \Pr[\mathbf{R}_0(\mathbf{Q}_0)[2] = \mathbf{U}\mathbf{Y}].$$

Proof. Write $\mathbf{U}$ as $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $\mathbf{X}^{-1}\mathbf{U}\mathbf{X} = \begin{pmatrix} a & \xi b \\ \xi^{-1}c & d \end{pmatrix}$ and $\mathbf{U}\mathbf{Y} = \begin{pmatrix} \xi a & \xi^{-1}b \\ \xi c & \xi^{-1}d \end{pmatrix}$.

Note

$$\begin{pmatrix} a & b \\ \xi^{-1}c & d \end{pmatrix}^* \mathbf{Q}_0 \begin{pmatrix} a & b \\ \xi^{-1}c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^* \mathbf{Q}_0 \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \xi a & \xi b \\ \xi c & \xi d \end{pmatrix}^* \mathbf{Q}_0 \begin{pmatrix} \xi a & \xi b \\ \xi c & \xi d \end{pmatrix}$$

and

$$\begin{pmatrix} a & b \\ \xi^{-1}c & d \end{pmatrix}^* \mathbf{Q}_0 \begin{pmatrix} \xi b \\ d \end{pmatrix} = \xi \begin{pmatrix} a & b \\ c & d \end{pmatrix}^* \mathbf{Q}_0 \begin{pmatrix} b \\ d \end{pmatrix}, \begin{pmatrix} \xi a & \xi b \\ \xi c & \xi d \end{pmatrix}^* \mathbf{Q}_0 \begin{pmatrix} \xi^{-1}b \\ \xi^{-1}d \end{pmatrix} = \xi^{-2} \begin{pmatrix} a & b \\ c & d \end{pmatrix}^* \mathbf{Q}_0 \begin{pmatrix} b \\ d \end{pmatrix}$$

since $\mathbf{Q}_0$ is diagonal. Also note the action of multiplying by ξ can be viewed as a sign permutation on the coefficients and then $\forall f \in \mathbb{L}$, $\lceil f \rceil = 0$ iff $\lceil \xi f \rceil = 0$ iff $\lceil \xi^{-2} f \rceil = 0$.

So we have $\rho_{\mathbf{Q}_0, \sigma} \left(\begin{pmatrix} a \\ c \end{pmatrix} \right) = \rho_{\mathbf{Q}_0, \sigma} \left(\begin{pmatrix} a \\ \xi^{-1}c \end{pmatrix} \right) = \rho_{\mathbf{Q}_0, \sigma} \left(\begin{pmatrix} \xi a \\ \xi c \end{pmatrix} \right)$ and $\begin{pmatrix} \xi b \\ d \end{pmatrix}$ is reduced with respect to $\begin{pmatrix} a \\ \xi^{-1}c \end{pmatrix}$ and $\mathbf{Q}_0$ iff $\begin{pmatrix} b \\ d \end{pmatrix}$ is reduced with respect to $\begin{pmatrix} a \\ c \end{pmatrix}$ and $\mathbf{Q}_0$ iff $\begin{pmatrix} \xi^{-1}b \\ \xi^{-1}d \end{pmatrix}$ is reduced with respect to $\begin{pmatrix} \xi a \\ \xi c \end{pmatrix}$ and $\mathbf{Q}_0$.

Thus, by Lemma 4.1, we obtain the desired conclusion. $\square$

Note that the matrices in $\text{GSL}_2(\mathcal{O}_{\mathbb{L}})$ can always be uniquely represented as a product of a matrix in $\text{SL}_2(\mathcal{O}_{\mathbb{L}})$ and a matrix of the form $\text{diag}(1, \xi)$. A natural construction for a re-randomized algorithm on $(\text{GSL}_2(\mathcal{O}_{\mathbb{L}}), [Q_0])$ is to sample the part of $\text{SL}_2(\mathcal{O}_{\mathbb{L}})$ using $\mathbf{R}_0$ and uniformly randomly sample the part of the form $\text{diag}(1, \xi)$. When $\mathbf{Q}_0$ is diagonal, we consider the equivalence class $[Q_0]$, so it can be directly used as the input for $\mathbf{R}_0$. We next prove that such a construction is indeed a re-randomized algorithm that satisfies the requirement.

Lemma 4.3 (Re-Randomized algorithm) Assume $\mathcal{D}_{[\mathbf{Q}_0]}$ is a samplable distribution on $[\mathbf{Q}_0]$ s.t. $Q_0 \in \text{Supp}(\mathcal{D}_{[\mathbf{Q}_0]})$. If $\mathbf{R}_0^{[\mathbf{Q}_0]}$ (resp. $\mathbf{R}_0^{\text{Supp}(\mathcal{D}_{[\mathbf{Q}_0]})}$) is efficient, then the randomized algorithm $\mathbf{R}$ described in Algorithm 3 is efficient, with the following property:

– For any

$$\mathbf{Q} \in [\mathbf{Q}_0] (\text{resp. } \text{Supp}(\mathcal{D}_{[\mathbf{Q}_0]})), \mathbf{V} \in \text{GSL}_2(\mathcal{O}_{\mathbb{L}})$$

s.t. $\mathbf{V}^* \mathbf{Q} \mathbf{V} \in [\mathbf{Q}_0] (\text{resp. } \text{Supp}(\mathcal{D}_{[\mathbf{Q}_0]}))$, the distribution of $\mathbf{V}^{-1} \cdot \mathbf{R}(\mathbf{Q})[2]$ is the same as that of $\mathbf{R}(\mathbf{V}^* \mathbf{Q} \mathbf{V})[2]$.

And so does also if we consider the induced group action for $\text{PGSL}_2(\mathcal{O}_{\mathbb{L}})$.

Proof. It's easy to see its efficiency if R_0 is efficient. We only need to prove the properties.

Let $\mathbf{V} \in \text{GSL}_2(\mathcal{O}_{\mathbb{L}})$ and $\mathbf{Q}, \mathbf{V}^* \mathbf{Q} \mathbf{V} \in [\mathbf{Q}_0]$ (resp. $\text{Supp}(\mathcal{D}_{[\mathbf{Q}_0]})$). For the first claim, we want to show that for any $\mathbf{U} \in \text{GSL}_2(\mathcal{O}_{\mathbb{L}})$,

$$\Pr[\mathbf{R}(\mathbf{V}^* \mathbf{Q} \mathbf{V})[2] = \mathbf{U}] = \Pr[\mathbf{V}^{-1} \mathbf{R}(\mathbf{Q})[2] = \mathbf{U}].$$

Firstly, we write them as $\mathbf{Q} = \mathbf{B}^* \mathbf{Q}_0 \mathbf{B}$, $\mathbf{V} = \mathbf{V}_0 \mathbf{V}_1$, $\mathbf{U} = \mathbf{U}_0 \mathbf{U}_1$ where $\mathbf{B}, \mathbf{V}_0, \mathbf{U}_0 \in \text{SL}_2(\mathcal{O}_{\mathbb{L}})$ and $\mathbf{V}_1$ (resp. $\mathbf{U}_1$) = $\begin{pmatrix} 1 & 0 \\ 0 & \det(\mathbf{V}) \end{pmatrix}$ (resp. $\begin{pmatrix} 1 & 0 \\ 0 & \det(\mathbf{U}) \end{pmatrix}$). And so we can write $\mathbf{V}^* \mathbf{Q} \mathbf{V} = (\mathbf{V}_1^{-1} \mathbf{B} \mathbf{V}_0 \mathbf{V}_1)^* (\mathbf{V}_1^{-1} \mathbf{B} \mathbf{V}_0 \mathbf{V}_1) \mathbf{Q}_0$. Note that $\mathbf{V}_1^{-1} \mathbf{B} \mathbf{V}_0 \mathbf{V}_1 \in \text{SL}_2(\mathcal{O}_{\mathbb{L}})$. From the construction of $\mathbf{R}$, we can see

$$\begin{aligned} \Pr[\mathbf{R}(\mathbf{V}^* \mathbf{Q} \mathbf{V})[2] = \mathbf{U}] &= \frac{1}{n} \cdot \Pr[\mathbf{R}_0(\mathbf{V}^* \mathbf{Q} \mathbf{V})[2] = \mathbf{U}_0] \\ &= \frac{1}{n} \cdot \Pr[\mathbf{R}_0(\mathbf{Q}_0)[2] = (\mathbf{V}_1^{-1} \mathbf{B} \mathbf{V}_0 \mathbf{V}_1) \mathbf{U}_0]. \end{aligned}$$

The last equality holds by [Lemma 4.1](#) since $\mathbf{V}_1^{-1} \mathbf{B} \mathbf{V}_0 \mathbf{V}_1 \in \text{SL}_2(\mathcal{O}_{\mathbb{L}})$ and

$$(\mathbf{V}_1^{-1} \mathbf{B} \mathbf{V}_0 \mathbf{V}_1)^* \mathbf{Q}_0 (\mathbf{V}_1^{-1} \mathbf{B} \mathbf{V}_0 \mathbf{V}_1) = \mathbf{V}^* \mathbf{Q} \mathbf{V}$$

(note that $\mathbf{Q}_0$ implies $(\mathbf{V}_1^{-1})^* \mathbf{Q}_0 \mathbf{V}_1^{-1} = \mathbf{Q}_0$). Similarly, we have

$$\begin{aligned} \Pr[\mathbf{V}^{-1} \mathbf{R}(\mathbf{Q})[2] = \mathbf{U}] &= \Pr[\mathbf{R}(\mathbf{Q})[2] = \mathbf{V} \mathbf{U}] \\ &= \frac{1}{n} \cdot \Pr[\mathbf{R}_0(\mathbf{Q})[2] = \mathbf{V}_0 \mathbf{V}_1 \mathbf{U}_0 \mathbf{V}_1^{-1}] \\ &= \frac{1}{n} \cdot \Pr[\mathbf{R}_0(\mathbf{Q}_0)[2] = \mathbf{B}(\mathbf{V}_0 \mathbf{V}_1 \mathbf{U}_0 \mathbf{V}_1^{-1})]. \end{aligned}$$

Denote $\mathbf{B}(\mathbf{V}_0 \mathbf{V}_1 \mathbf{U}_0 \mathbf{V}_1^{-1})$ by $\mathbf{M} \in \text{SL}_2(\mathcal{O}_{\mathbb{L}})$, then

$$(\mathbf{V}_1^{-1} \mathbf{B} \mathbf{V}_0 \mathbf{V}_1) \mathbf{U}_0 = \mathbf{V}_1^{-1} \mathbf{M} \mathbf{V}_1.$$

By [Lemma 4.2](#),

$$\Pr[\mathbf{R}_0(\mathbf{Q}_0)[2] = \mathbf{M}] = \Pr[\mathbf{R}_0(\mathbf{Q}_0)[2] = \mathbf{V}_1^{-1} \mathbf{M} \mathbf{V}_1],$$

so

$$\Pr[\mathbf{R}(\mathbf{V}^* \mathbf{Q} \mathbf{V})[2] = \mathbf{U}] = \Pr[\mathbf{V}^{-1} \mathbf{R}(\mathbf{Q})[2] = \mathbf{U}].$$

For the second claim, we just need to show that for any $\mathbf{U} \in \text{GSL}_2(\mathcal{O}_{\mathbb{L}})$,

$$\sum_{\xi \in \mu(\mathbb{L})} \Pr[\mathbf{R}(\mathbf{V}^* \mathbf{Q} \mathbf{V})[2] = \xi I_2 \cdot \mathbf{U}] = \sum_{\xi \in \mu(\mathbb{L})} \Pr[\mathbf{V}^{-1} \mathbf{R}(\mathbf{Q})[2] = \xi I_2 \cdot \mathbf{U}],$$

which can be directly derived from the previous conclusion. $\square$

Remark 6. We can prove $\Pr[\mathbf{R}(\mathbf{V}^* \mathbf{Q} \mathbf{V})[2] = \mathbf{U}] = \Pr[\mathbf{R}(\mathbf{V}^* \mathbf{Q} \mathbf{V})[2] = \xi I_2 \cdot \mathbf{U}]$ for any $\xi \in \mu(\mathbb{L})$ by the other equality in [Lemma 4.2](#).

Algorithm 3: R: Re-Randomization for $(\text{GSL}_2(\mathcal{O}_{\mathbb{L}}),)$

Require: Conductor $m = 2^{\kappa}$ cyclotomic $\mathbb{L}$, $\mathbf{Q} \in \mathcal{H}_2(\mathbb{L})$
Ensure: $\mathbf{R} \in [\mathbf{Q}]_{\text{gsl}}$ and $\mathbf{U} \in \text{GSL}_2(\mathcal{O}_K)$ such that $\mathbf{R} = \mathbf{U}^* \cdot \mathbf{Q} \cdot \mathbf{U}$
1: Let $(\mathbf{R}_0, \mathbf{U}_0) \leftarrow \text{R}_0(\mathbf{Q})$
2: Parse $\xi \leftarrow_{\S} \mu(\mathbb{L})$
3: Let $Y = \begin{pmatrix} 1 & 0 \\ 0 & \xi \end{pmatrix}$, $\mathbf{U} = \mathbf{U}_0 Y$, and $\mathbf{R} = \mathbf{U}^* \mathbf{Q} \mathbf{U}$
4: **return** $(\mathbf{R}, \mathbf{U})$

4.3 Extractor on $\text{PGSL}_2(\mathcal{O}_{\mathbb{L}})$ **Lemma 4.4** *For the group actions*

$$(\text{PGSL}_2(\mathcal{O}_{\mathbb{L}}), [\mathbf{Q}_{\mathbf{k}}^{(0)}]_{\text{gsl}}, \star), (\text{PGSL}_2(\mathcal{O}_{\mathbb{L}}), [\mathbf{Q}_{\ell}^{(2)}]_{\text{gsl}}, \star), k \geq 0, \ell > 2$$

and the group $M = (\langle X \rangle / \langle X^2 \rangle, \times) \simeq (\{\pm 1\}, \times)$, define $\mathbf{E} : \text{PGSL}_2(\mathcal{O}_{\mathbb{L}}) \rightarrow M$ such that $\mathbf{E}(\mathbf{U}/\mu(\mathbb{L})I_2) = \det(\mathbf{U})/\langle X^2 \rangle$. Then, $\mathbf{E}$ is an extractor satisfying all properties listed in [Definition 2.11](#).

Proof. Homomorphic property: It's easy to see $\mathbf{E}$ is a well-defined surjective group homomorphism (note $\mathbf{E}(\mu(\mathbb{L})I_2) = \langle X^2 \rangle$). Uniformly extracting property on $[\mathbf{Q}_{\mathbf{k}}^{(0)}]_{\text{gsl}}$: For any $\mathbf{Q} \in [\mathbf{Q}_{\mathbf{k}}^{(0)}]_{\text{gsl}}$,

$$\mathbf{E}(\text{Stab}(\mathbf{Q})) = \det(\text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q}))/\langle X^2 \rangle = \det(\text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q}_{\mathbf{k}}^{(0)}))/\langle X^2 \rangle = M.$$

Thus, $\mathbf{E}$ is surjective even when restricted to the subgroup $\text{Stab}(\mathbf{Q})$. Furthermore, $\mathbf{E}(U(\text{Stab}(\mathbf{Q}))) = U(M)$ from the invariance of the $U(\text{Stab}(\mathbf{Q}))$ to shifts by elements from that same group. Then for any $\mathbf{Q}, \mathbf{Q}' \in [\mathbf{Q}_{\mathbf{k}}^{(0)}]_{\text{gsl}}$, assume $\mathbf{Y} \in \mathcal{I}(\mathbf{Q}, \mathbf{Q}')$. We have

$$\mathbf{E}(U(\mathcal{I}(\mathbf{Q}, \mathbf{Q}')))) = \mathbf{E}(\mathbf{Y} \cdot U(\text{Stab}(\mathbf{Q}))) = \mathbf{E}(\mathbf{Y}) \cdot U(M) = U(M).$$

Local constant property on $[\mathbf{Q}_{\mathbf{k}}^{(2)}]_{\text{gsl}}$: For any $\mathbf{Q} \in [\mathbf{Q}_{\mathbf{k}}^{(2)}]_{\text{gsl}}$, $\mathbf{E}(\text{Stab}(\mathbf{Q})) = \det(\text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q}_{\mathbf{k}}^{(2)}))/\langle X^2 \rangle$ is trivial by [Lemma 3.2](#). Since $\mathbf{E}$ is a homomorphism, for any $\mathbf{Q}_1, \mathbf{Q}_2 \in [\mathbf{Q}_{\mathbf{k}}^{(2)}]_{\text{gsl}}$, $\mathbf{E}(\mathcal{I}(\mathbf{Q}_1, \mathbf{Q}_2))$ is a constant.

Trapdoor property on $[\mathbf{Q}_{\mathbf{k}}^{(0)}]_{\text{gsl}}$: By the argument presented in [Lemma 2.8](#), the result follows. $\square$

A Discussion on the extractor. Previously, we only briefly mentioned that because the $\det$ cannot serve as an extractor on $\text{SL}_2(\mathcal{O}_{\mathbb{L}})$, additional efforts are required to adjust the group and the re-randomized algorithm. In fact, if we wish to preserve *the uniformly extracting property and homomorphic property*, it can be proven that for the equivalence classes we consider, there exists no extractor meeting the requirements if we select $\text{SL}_2(\mathcal{O}_{\mathbb{L}})$ for the group action. This proof will be provided in the [Appendix D](#).

4.4 Commitments

The Enhanced Linkable Commitment We take group action $(\text{PGSL}_2(\mathcal{O}_{\mathbb{L}}), [\mathbf{Q}_k^{(0)}]_{\text{gs1}}, \star)$ with constant k , and a samplable distribution $\mathcal{D}_{[\mathbf{Q}_k^{(0)}]_{\text{gs1}}}$, under [Heuristic 1](#) for $\mathbf{Q}_k^{(0)}$, by [Lemma 4.3](#), [Lemma 4.4](#), [Theorem 2.2](#) and [Theorem 3.1](#), there is an Enhanced Linkable Commitment based on the hardness of module-LIP $_{\mathcal{D}_{[\mathbf{Q}_k^{(0)}]_{\text{gs1}}}}$. A natural choice of $\mathcal{D}_{[\mathbf{Q}_k^{(0)}]_{\text{gs1}}}$ is to take it as $\text{R}(\mathbf{Q}_k^{(0)})[1]$ with $\sigma = 2^{\Theta(n)}$, in which case the security of the resulting commitment is based on the worst-case hardness of the module-LIP $_{[\mathbf{Q}_k^{(0)}]}$. And module-LIP $_{[\mathbf{Q}_k^{(0)}]}$ is as hard as $\mathcal{O}_{\mathbb{L}}^2$ -LIP by [Theorem 3.3](#).

When $k = 0$, another choice of the distribution $\mathcal{D}_{[I_2]}$ is using the public key distribution from HAWK, the resulting commitment scheme inherits the secret key security of HAWK.

Dual-Mode Commitment We take group actions $(\text{PGSL}_2(\mathcal{O}_{\mathbb{L}}), [\mathbf{Q}_{2k}^{(0)}]_{\text{gs1}}, \star)$, $(\text{PGSL}_2(\mathcal{O}_{\mathbb{L}}), [\mathbf{Q}_k^{(2)}]_{\text{gs1}}, \star)$ with $1 < k = O(\log n)$, and samplable distributions $\mathcal{D}_{[\mathbf{Q}_k^{(0)}]_{\text{gs1}}}$ and $\mathcal{D}_{[\mathbf{Q}_k^{(0)}]_{\text{gs1}}}$, under [Heuristic 1](#) for $\mathbf{Q}_{2k}^{(0)}$, by [Lemma 4.3](#), [Lemma 4.4](#), and [Theorem 2.2](#), there is a Dual-Mode Commitment based on the hardness of module-LIP $_{\mathcal{D}_{[\mathbf{Q}_{2k}^{(0)}]_{\text{gs1}}}}$ and module- $\Delta\text{LIP}_{\mathbb{L}, \mathcal{D}_{[\mathbf{Q}_{2k}^{(0)}]_{\text{gs1}}}, \mathcal{D}_{[\mathbf{Q}_k^{(2)}]_{\text{gs1}}}}$.

When taking $\mathcal{D}_{[\mathbf{Q}_{2k}^{(0)}]_{\text{gs1}}}$ as $\text{R}(\mathbf{Q}_{2k}^{(0)})[1]$ and $\mathcal{D}_{[\mathbf{Q}_k^{(2)}]_{\text{gs1}}}$ as $\text{R}(\mathbf{Q}_k^{(2)})[1]$ with $\sigma = 2^{\Theta(n)}$, the hardness of module- $\Delta\text{LIP}_{\mathbb{L}, \mathcal{D}_{[\mathbf{Q}_{2k}^{(0)}]_{\text{gs1}}}, \mathcal{D}_{[\mathbf{Q}_k^{(2)}]_{\text{gs1}}}}$ achieves that of the worst-case module- $\Delta\text{LIP}_{\mathbb{L}, [\mathbf{Q}_{2k}^{(0)}], [\mathbf{Q}_k^{(2)}]}$, which was discussed in Section 3.4.

4.5 Discussion about Asymptotic Complexity

Plain version v.s. Module version. Since the main body of all schemes is based on either the basic scheme or the dual scheme corresponding to [Protocol 2.1](#), and the plain version did not provide concrete instance selection for the dual scheme, we focus our analysis on the asymptotic complexity of the basic scheme.

The process of generating Ck involves sampling from $\mathcal{D}_{[\mathbf{Q}_k^{(0)}]}$ and $\mathcal{D}_{[I_{2n}]}$ (here k is a small constant). Assuming the hardness of module-LIP holds when the entry size is bounded by $B = O(\log n)$: In the **module version**, following a key generation approach similar to HAWK, the time complexity is $O(n \log n)$, and the size of Ck is $O(n \log n)$. In the **plain version**, it requires sampling several vectors in $\mathbb{Z}^n$, computing the HNF of a matrix in $\mathbb{Z}^{2n \times 2n}$, and performing two matrix multiplications. The size of Ck is at least $O(n^2 \log n)$.

For the commitment generation process: The **module version** requires a constant number of discrete Gaussian samplings on Ck with $\sigma = 2^{O(\log n)}$, **HermiteSolve**, and element multiplications/inversions in $\mathbb{L}$ with polynomial coefficients of size $O(n \log n)$. The commitment size is $O(n^2 \log n)$. The **plain version** requires $O(n^2)$ discrete Gaussian samplings on Ck with $\sigma = 2^{O(\log n)}$, one

HNF computation, one determinant calculation, and two matrix multiplications. The commitment size is $O(n^3 \log n)$.

For the opening process: The **module version** requires a constant number of element multiplications in $\mathbb{L}$ with polynomial coefficients of size $O(n \log n)$. The **plain version** requires two matrix multiplications with entries of size $O(n \log n)$ and one determinant computation.

It can be observed that the module version demonstrates advantages over the plain version. In fact, the element size inflation primarily stems from the HNF computation. By leveraging the properties of **Reduce**, during the commitment generation process, one can reject sampled vectors that satisfy $\langle (x^* \mathbf{Q}x)^{-1}, 1 \rangle > \text{poly}(n)$, thereby maintaining the final commitment size at $O(n \log n)$. Consequently, the polynomial coefficients in the opening process remain of size $O(\log n)$. It can be proven that this rejection sampling does not affect the properties of the re-randomized algorithm. We omit a detailed explanation of this aspect, as the excessively heavy HNF computation step still renders the overall scheme impractical.

Table 1. Complexity and Size Comparison between Plain and Module Schemes

Schemes	Time Complexity			Size	
	Gen (1^λ)	Com _{y} (m)	Open _{y} (c, d)	Ck	Commitment
Plain	T_{HNF}	$n^2 T_{\text{DGS}} + T_{\text{HNF}}$	$\tilde{O}(n^{\omega+1} \log n)$	$O(n^2 \log n)$	$O(n^3 \log n)$
Module	$\tilde{O}(n \log n)$	T_{HNF}	$\tilde{O}(n^2 \log n)$	$O(n \log n)$	$O(n^2 \log n)$

Note: Let T_{HNF} and T_{DGS} denote the asymptotic complexities of computing the HNF for a $2n \times 2n$ matrix and performing $2n$ -dimensional discrete Gaussian sampling, respectively. The term $O(n^\omega)$ represents the complexity of matrix multiplication. The notation $\tilde{O}$ indicates the omission of $\log \log n$ factors.

HermiteSolve v.s. TowerSolve. The most computationally heavy component in our scheme is the **HermiteSolve** used in the re-randomized algorithm to generate matrices in $\text{SL}_2(\mathcal{O}_{\mathbb{L}})$, which involves computing the Hermite Normal Form (HNF) of integer matrices. HAWK also requires this operation [10]. In their practical scheme, they opt to replace **HermiteSolve** with **TowerSolve** from [28], which has a complexity of only $O(n \log(\|f\| + \|g\|))$ but imposes additional requirements on the input. To justify this, they conducted experiments and provided theoretical arguments, heuristically suggesting that the inputs solvable by **TowerSolve** approximate those solvable by **HermiteSolve**.

For simplicity, we focus only on the basis scheme constructed from $\mathbf{Q}_k^{(0)}$. We also wish to make a similar substitution, even though this would directly compromise the properties of the re-randomized algorithm **R** (denoting the substituted version as **R'**). The primary purpose of the current re-randomized algorithm is to achieve perfect hiding. After substitution, hiding may still be computationally hard—that is, although the output distribution of **R'** is not uniform over

$\text{Aut}_{\mathcal{O}_L}(\mathbf{Q})$, there is no evidence suggesting that any information about $\text{Aut}_{\mathcal{O}_L}(\mathbf{Q})$ can be extracted from $\mathbf{Q}$. However, we have not been able to reduce this to any known problem.

When considering even $k > 2$, if we assume that $\text{module-}\Delta\text{LIP}_{[\mathbf{Q}_k^{(0)}], [\mathbf{Q}_{k/2}^{(2)}]}$ (under an appropriate distribution) is hard, we can reconsider the hardness from another perspective. This is because the computational hiding hardness of the basis scheme based on $\mathbf{Q}_k^{(0)}$ is equivalent to that of the dual scheme based on $\mathbf{Q}_{k/2}^{(2)}$ (if an adversary has an advantage in the hiding game for only one scheme, then by conducting a hiding game on a given Hermitian form and invoking the adversary, we can determine which isomorphism class the given Hermitian form belongs to based on whether an advantage exists). The hiding hardness of the dual scheme is more intuitive because the value extracted by the extractor $\mathbf{E}$ from $\text{Aut}_{\mathcal{O}_L}(\mathbf{Q})$ is inherent to $\mathbf{Q}$. That is, for a $\mathbf{Q}$ in $[\mathbf{Q}_{k/2}^{(2)}]$, the goal is to determine the value of $\mathbf{E}(\text{Aut}_{\mathcal{O}_L}(\mathbf{Q}))$. Its worst-case is hard, and there exists a distribution over $[\mathbf{Q}_{k/2}^{(2)}]$ that implies worst-case hardness (the output distribution of original $\mathbf{R}$ in the dual scheme). We hope that after some subtle rejection sampling, this distribution also implies hardness.

5 Conclusion

This paper first theoretically investigates the hardness of specific module-LIP problems and then, building on this security foundation, utilizes the framework proposed in [16] to construct group-action-based commitment schemes in the module setting by selecting appropriate group actions. Compared to the original plain setting, these schemes achieve efficiency improvements. Although not yet practical, their module structure provides optimization potential for practical implementations.

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Appendix A Supplementary Definitions for Commitment Schemes

Definition A.1 (Enhanced Linkable Commitment) *Let λ denote the security parameter, $(M, \cdot)$ represent a finite group serving as the message space, and (C, D) denote the commitment and opening spaces respectively. An enhanced linkable commitment scheme is a tuple $\Pi_{elc} = (\mathbf{Gen}, \mathbf{Com}, \mathbf{Open}, (\mathbf{Link}, \mathbf{LinkE}, \mathbf{LinkC}))$, consisting of a commitment scheme Π_{com} and a link component. The scheme satisfies the following properties:*

1. *The link component comprises PPT algorithms $(\mathbf{Link}, \mathbf{LinkE}, \mathbf{LinkC})$. For any two messages m_0 and m_1 in M with commitments $(c_0, d_0) = \mathbf{Com}(m_0)$ and $(c_1, d_1) = \mathbf{Com}(m_1)$:*
 - $\mathbf{Link}(c_0, c_1, d_L)$ returns 0 or 1. If $\mathbf{Link}(c_0, c_1, d_L) = 1$, d_L is a valid linking value for the pair (c_0, c_1) .
 - $\mathbf{LinkE}(d_0, d_1)$ generates a linking value d_L such that $\mathbf{Link}(c_0, c_1, d_L) = 1$.

– $\text{LinkC}(c_0, c_1, d_L)$ computes $m_0 \cdot m_1^{-1}$ if $\text{Link}(c_0, c_1, d_L) = 1$, else returns $\perp$.

2. **Computationally Enhanced-Linkable-Hiding (ELH)**: No PPT adversary $\mathcal{A}$ can win the **ELH** game with probability exceeding $\frac{1}{2} + \text{negl}(\lambda)$:

$$\Pr[\mathcal{A} \text{ wins } \mathbf{ELH}(\Pi_{elc})] \leq \frac{1}{2} + \text{negl}(\lambda).$$

Here, $\mathbf{ELH}(\Pi_{elc})$ represents the game described in Figure 1.

3. **Computationally Enhanced-Linkable-Binding (ELB)**: No PPT adversary $\mathcal{A}$ can violate the binding property with probability higher than $\text{negl}(\lambda)$:

$$\Pr \left[\begin{array}{l} \mathbf{Open}(c_i, d_i) = m_i, i \in \{0, 1\}, \\ \text{Link}(c_0, c_1, d_L) = 1, \\ m_0 \cdot m_1^{-1} \neq \text{LinkC}(c_0, c_1, d_L) \end{array} \middle| \begin{array}{l} Ck \leftarrow \mathbf{Gen}(1^\lambda), \\ (m_0, m_1, c_0, c_1) \leftarrow \mathcal{A}(Ck) \\ d_0, d_1, d_L \end{array} \right] \leq \text{negl}(\lambda).$$

4. **Computationally Enhanced-Linkable-Unforgeable (ELU)**: No PPT adversary $\mathcal{A}$ can win the **ELU** game with probability exceeding $\text{negl}(\lambda)$:

$$\Pr[\mathcal{A} \text{ wins } \mathbf{ELU}(\Pi_{elc})] \leq \text{negl}(\lambda).$$

Here, $\mathbf{ELU}(\Pi_{elc})$ represents the game described in Figure 2.

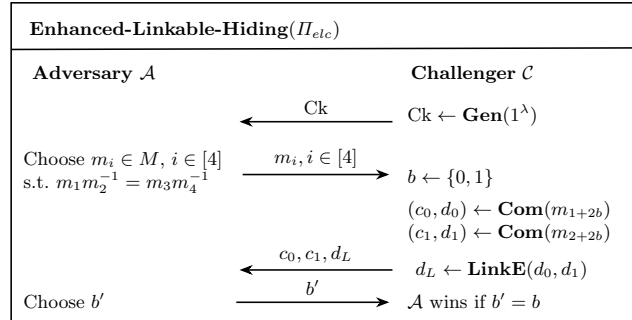
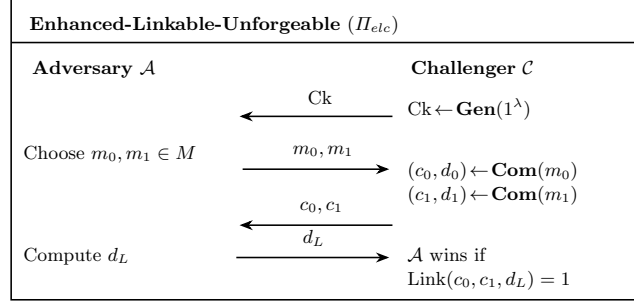


Fig. 1. The Enhanced Linkable Hiding Game

**Fig. 2.** The Enhanced Linkable Unforgeable Game

Commitment Scheme A.1 : $\Pi_{elc} = (\mathbf{Gen}, \mathbf{Com}, \mathbf{Open}, (\mathbf{Link}, \mathbf{LinkE}, \mathbf{LinkC}))$, for a security parameter λ .

– $\Pi_{com} = (\mathbf{Gen}, \mathbf{Com}, \mathbf{Open})$ is [Protocol 2.1](#).

If $Ck = y$, for two commitments of m_0 and m_1 in [Protocol 2.1](#):

$$\begin{aligned} \mathbf{Com}_y(m_0) &\rightarrow (c_0, d_0) = ((c_{01}, c_{02}), d_0) = ((E(g_0) \cdot m_0, y_0), g_0) \\ \mathbf{Com}_y(m_1) &\rightarrow (c_1, d_1) = ((c_{11}, c_{12}), d_1) = ((E(g_1) \cdot m_1, y_1), g_1) \end{aligned}$$

The link component $(\mathbf{Link}, \mathbf{LinkE}, \mathbf{LinkC})$ as follows:

- $\mathbf{Link}(c_0, c_1, d_L) = \begin{cases} 1 & \text{if } d_L \star c_{12} = c_{02}. \\ 0 & \text{else.} \end{cases}$
- $\mathbf{LinkE}(d_0, d_1) = d_0 \cdot d_1^{-1}$.
- $\mathbf{LinkC}(c_0, c_1, d_L)$: If $d_L \star c_{12} = c_{02}$, it outputs $E(d_L)^{-1} \cdot c_{01} \cdot c_{11}^{-1}$; otherwise, it outputs $\perp$.

Definition A.2 (Dual-Mode Commitment) A dual-mode commitment scheme is a tuple of PPT algorithms $\Pi_{dmc} = (\mathbf{Gen}, \mathbf{TGen}, \mathbf{Com}, \mathbf{TCom}, \mathbf{Open}, \mathbf{TCol})$, for a security parameter λ , message space M , commitment space C , and opening space D , such that

1. $(\mathbf{TGen}, \mathbf{TCom}, \mathbf{TCol})$ satisfies:
 - $\mathbf{TGen}(1^\lambda)$ outputs the public commitment key Ck and corresponding trapdoor information Td .
 - $\mathbf{TCom}(Ck, Td)$ outputs a committing value $c \in C$ and a state St .
 - $\mathbf{TCol}(Ck, Td, St, m)$ outputs an opening value $d \in D$.
2. **Perfectly Binding:** $(\mathbf{Gen}, \mathbf{Com}, \mathbf{Open})$ is a perfectly binding commitment scheme.
3. **Completeness:** For $(Ck, Td) \leftarrow \mathbf{TGen}$ and $m \in M$, it holds that

$$\Pr[\mathbf{Open}_{Ck}(c, d) = m \mid (c, d) \leftarrow \mathbf{Com}_{Ck}(m)] = 1.$$

4. **Trapdoor Property:** For $(Ck, Td) \leftarrow \mathbf{TGen}$ and $m \in M$, the following distributions are identical:

$$\{(c, d, m) \mid (c, d) \leftarrow \mathbf{Com}_{Ck}(m)\} \quad \text{and}$$

$$\{(c, d, m) \mid (c, St) \leftarrow \mathbf{TCom}(Ck, Td), d \leftarrow \mathbf{TCol}(Ck, Td, St, m)\}.$$

5. **Key Indistinguishability:** For all PPT adversaries $\mathcal{A}$, there exists a negligible function $\text{negl}(\lambda)$ such that

$$\Pr \left[b = \tilde{b} \mid \begin{array}{l} Ck_0 \leftarrow \mathbf{Gen}(1^\lambda), Ck_1 \leftarrow \mathbf{TGen}(1^\lambda) \\ b \leftarrow \{0, 1\}, \tilde{b} \leftarrow \mathcal{A}(Ck_b) \end{array} \right] \leq \frac{1}{2} + \text{negl}(\lambda).$$

Commitment Scheme A.2 : $\Pi_{dmc} = (\mathbf{Gen}, \mathbf{TGen}, \mathbf{Com}, \mathbf{TCom}, \mathbf{Open}, \mathbf{TCol})$, for a security parameter λ .

- $\mathbf{Gen}(1^\lambda)$: Sample $y \leftarrow \mathcal{D}_{X'}$ and output $Ck = y$.
- $\mathbf{TGen}(1^\lambda)$: Sample $y = g \star x \leftarrow \mathcal{D}_X$ and compute $\text{Stab}(y) = g \cdot \text{Stab}(x) \cdot g^{-1}$. Output $(Ck, Td) = (y, \text{Stab}(y))$.
- $\mathbf{Com}_y(m)$: For a message $m \in M$, sample $(y', g) \leftarrow \mathbf{R}(y)$ such that $g \star y = y'$. Compute $c = (c_1, c_2) = (\mathbf{E}(g) \cdot m, y')$ and set $d = g$. Output (c, d) .
- $\mathbf{TCom}(y, \text{Stab}(y))$: Sample $(y', g) \leftarrow \mathbf{R}(y)$, $u \leftarrow M$ and set $c = (c_1, c_2) = (u, y')$, $St = (u, g)$. Output (c, St) .
- $\mathbf{Open}_y(c, d)$: If $d \star y = c_2$, output $\tilde{m} = \mathbf{E}(d)^{-1} \cdot c_1$; otherwise, $\tilde{m} = \perp$.
- $\mathbf{TCol}(y, \text{Stab}(y), (u, g), m)$: Compute $r = u \cdot m^{-1}$, then sample $\mathbf{F}(r, g, y, \text{Stab}(y)) \rightarrow g'$ and set $d = g'$. Output d .

Appendix B Generalized Theorem 3.1 for CM number field

The main structure of the reduction is identical to Theorem 3.1. What requires additional proof is that for a general CM field $\mathbb{L}$, after considering its extension of the form $\mathbb{L}[Y]/(Y^2 - \xi)$ (where ξ is a root of unity), the extended field remains a CM field and an integral basis of its ring of integers can still be efficiently computed.

Lemma B.1 Assume $\xi \in \mathbb{L}$ is a root of unity and $X^2 - \xi$ doesn't have roots in $\mathbb{L}$. Let $\mathbb{F} := \mathbb{L}[X]/(X^2 - \xi)$. Then $\mathbb{F}$ is also a CM number field.

Proof. Let $\mathbb{K}$ be the totally real number field such that of $\mathbb{L}|\mathbb{K}$ is a totally imaginary quadratic extension. If $\xi \neq -1$, consider $\mathbb{M} := \mathbb{K}(X + X^{-1}) \subseteq \mathbb{F}$. $\mathbb{M}$ is totally real since $\mathbb{Q}(X + X^{-1})$ and $\mathbb{K}$ are both totally real. And $\mathbb{M} \neq \mathbb{K}$, otherwise

$$(\xi + 1)/X = (X^2 + 1)/X = X + X^{-1} \in \mathbb{K} \subseteq \mathbb{L} \Rightarrow X \in \mathbb{L},$$

leading to a contradiction.

If $\xi = -1$, assume $\mathbb{L} = \mathbb{K}(\alpha)$ in which $\alpha^2 \in \mathbb{K}$ and $\phi(\alpha^2) < 0$ for any embedding $\phi : \mathbb{K} \rightarrow \mathbb{C}$. Consider $\mathbb{M} := \mathbb{K}(X \cdot \alpha) \subseteq \mathbb{F}$. M is totally real since $\mathbb{K}$ is totally real and $\phi((X \cdot \alpha)^2) = -\phi(\alpha^2) > 0$ for any embedding $\phi : \mathbb{M} \rightarrow \mathbb{C}$. And $M \neq \mathbb{K}$, otherwise $X \cdot \alpha \in \mathbb{K} \subseteq \mathbb{L} \Rightarrow X \in \mathbb{L}$, leading to a contradiction. $\square$

When $\mathbb{L}$ is a totally real number field and $\mathbb{F} = \mathbb{L}[X]/(X^2 + 1)$, Mureau et.al. showed $\log \Delta_{\mathbb{F}} = \text{poly}(\log \Delta_{\mathbb{L}})$ and we can compute $\mathcal{O}_{\mathbb{F}}$ from $\mathcal{O}_{\mathbb{L}}$ in [26, Section 2.2]. We generalize this conclusion as follows.

Lemma B.2 ([5, Lemma 1.4]) *There are polynomial time algorithms that given an algebraic number field K and one of (a), (b), determine the other:*

- (a) *the ring of algebraic integers of K ;*
- (b) *the largest squarefree divisor of the discriminant of K .*

Lemma B.3 *Assume $\xi \in \mathbb{L}$ is a root of unity and $X^2 - \xi$ doesn't have roots in $\mathbb{L}$. Let $\mathbb{F} := \mathbb{L}[X]/(X^2 - \xi)$. There exists a polynomial time algorithm A that, given as input a $\mathbb{Z}$ -basis $B_{\mathbb{L}}$ of $\mathcal{O}_{\mathbb{L}}$, computes a $\mathbb{Z}$ -basis $B_{\mathbb{F}}$ of $\mathcal{O}_{\mathbb{F}}$.*

Proof. One can see $A := \mathcal{O}_{\mathbb{L}} + \mathcal{O}_{\mathbb{L}} \cdot X \subseteq \mathcal{O}_{\mathbb{F}}$ is an order of $\mathbb{F}$. Assume $B_{\mathbb{L}} = \{\beta_1, \dots, \beta_n\}$, then $\Delta_{\mathbb{L}} = \det((\text{Tr}_{\mathbb{L}}(\beta_i \beta_j)))$ and A 's discriminant over $\mathbb{Z}$ is

$$\Delta_A = \det \left(\begin{pmatrix} (\text{Tr}_{\mathbb{F}}(\beta_i \beta_j)) & (\text{Tr}_{\mathbb{F}}(X \beta_i \beta_j)) \\ (\text{Tr}_{\mathbb{F}}(X \beta_j \beta_i)) & (\text{Tr}_{\mathbb{F}}(\xi \beta_i \beta_j)) \end{pmatrix} \right).$$

On one hand, $A \subseteq \mathcal{O}_{\mathbb{F}}$ implies $\Delta_{\mathbb{F}} | \Delta_A$. On the other hand, we compute Δ_A to show it's just $2^{2n} \Delta_{\mathbb{L}}^2$. Thus $\Delta_{\mathbb{F}} | 2^{2n} \Delta_{\mathbb{L}}^2$. And we also know $\Delta_{\mathbb{L}} | \Delta_{\mathbb{F}}$. So the largest squarefree divisor of $\Delta_{\mathbb{F}}$ is the largest squarefree divisor of $\Delta_{\mathbb{L}}$ or the largest squarefree divisor of $\Delta_{\mathbb{L}}$ multiplied by 2. By Lemma B.2, we can find the largest squarefree divisor of $\Delta_{\mathbb{L}}$ from $B_{\mathbb{L}}$. Then we can guess the largest squarefree divisor of $\Delta_{\mathbb{F}}$ and compute the $\mathbb{Z}$ -basis of $\mathcal{O}_{\mathbb{F}}$ by Lemma B.2.

The computation is as following. Note that

$$\begin{aligned} \text{Tr}_{\mathbb{F}}(X \beta_i \beta_j) &= \text{Tr}_{\mathbb{L}}(\text{Tr}_{\mathbb{F}/\mathbb{L}}(X \beta_i \beta_j)) = \text{Tr}_{\mathbb{L}}(0) = 0, \\ \text{Tr}_{\mathbb{F}}(\beta_i \beta_j) &= \text{Tr}_{\mathbb{L}}(\text{Tr}_{\mathbb{F}/\mathbb{L}}(\beta_i \beta_j)) = 2\text{Tr}_{\mathbb{L}}(\beta_i \beta_j), \\ \text{Tr}_{\mathbb{F}}(\xi \beta_i \beta_j) &= \text{Tr}_{\mathbb{L}}(\text{Tr}_{\mathbb{F}/\mathbb{L}}(\xi \beta_i \beta_j)) = 2\text{Tr}_{\mathbb{L}}(\xi \beta_i \beta_j). \end{aligned}$$

Thus we have

$$\Delta_A = \det((2\text{Tr}_{\mathbb{L}}(\beta_i \beta_j))) \cdot \det((2\text{Tr}_{\mathbb{L}}(\xi \beta_i \beta_j))) = 2^{2n} \Delta_{\mathbb{L}} \cdot \det((\text{Tr}_{\mathbb{L}}(\xi \beta_i \beta_j))).$$

For $(\text{Tr}_{\mathbb{L}}(\xi \beta_i \beta_j))$, write it as $(\sigma_j(\xi \beta_i)) \cdot (\sigma_i(\beta_j))$, where $\{\sigma_i\}$ is $\text{Hom}_{\mathbb{Q}}(\mathbb{L}, \mathbb{C})$. Then

$$\begin{aligned} \det((\text{Tr}_{\mathbb{L}}(\xi \beta_i \beta_j))) &= \det((\sigma_j(\xi \beta_i))) \cdot \det((\sigma_i(\beta_j))) \\ &= \text{Nm}_{\mathbb{L}}(\xi) \cdot \det((\sigma_j(\beta_i))) \cdot \det((\sigma_i(\beta_j))) \\ &= \det((\sigma_j(\beta_i))) \cdot \det((\sigma_i(\beta_j))) \\ &= \det((\text{Tr}_{\mathbb{L}}(\beta_i \beta_j))) = \Delta_{\mathbb{L}}. \end{aligned}$$

$\square$

From the proof, we can see $\Delta_{\mathbb{F}} \leq 2^{2n} \Delta_{\mathbb{L}}^2$. Note $n = \text{poly}(\log \Delta_{\mathbb{L}})$, so $\log \Delta_{\mathbb{F}} = \text{poly}(\log \Delta_{\mathbb{L}})$.

Appendix C Omitted Proofs

Proof (of Lemma 2.7). If R has the property: for any $x, x' \in X$ (resp. $\text{Supp}(\mathcal{D}_X)$), $h \in \mathcal{I}(x, x')$, the distributions of $R(h \star x)[2]$ and $R(x)[2] \cdot h^{-1}$ are same. This means for any $x, x' \in X$ (resp. $\text{Supp}(\mathcal{D}_X)$), $h \in \mathcal{I}(x, x')$, $g \in G$, $\Pr[R(h \star x)[2] = gh^{-1}] = \Pr[R(x)[2] \cdot h^{-1} = gh^{-1}] = \Pr[R(x)[2] = g]$.

For any $x, x' \in X$ (resp. $\text{Supp}(\mathcal{D}_X)$), and $(x'', g) \leftarrow R(x)$, assume $h \in \mathcal{I}(x, x')$. Firstly,

$$\begin{aligned} \Pr[R(x)[1] = x''] &= \Pr[R(x)[2] \in \mathcal{I}(x, x'')] \\ &= \sum_{f \in \mathcal{I}(x, x'')} \Pr[R(x)[2] = f] \\ &= \sum_{f \in \mathcal{I}(x, x'')} \Pr[R(h \star x)[2] = f \cdot h^{-1}] \\ &= \sum_{t \in \mathcal{I}(x', x'')} \Pr[R(x')[2] = t] = \Pr[R(x')[1] = x'']. \end{aligned}$$

The third equality uses the property, and the fourth equality uses $\mathcal{I}(x', x'') = \mathcal{I}(x, x'') \cdot h^{-1}$.

Secondly, for any $g, g' \in \mathcal{I}(x, x'')$, we have

$$\Pr[R(x)[2] = g'] = \Pr[R((g^{-1} \cdot g') \star x)[2] = g' \cdot (g^{-1} \cdot g')^{-1}] = \Pr[R(x)[2] = g].$$

□

Proof (of Theorem 2.1). For an $\mathcal{O}_{\mathbb{L}}^2$ -LIP instance $\mathbf{Q} = U^*U$, select v_1 from $\{(1, 0)^T, (0, 1)^T, (1, 1)^T\}$ such that $U^{-1}(J_2 t_* u_1 - u_1) = U^{-1} J_2 t_* U v_1 - v_1$ does not belong to $\langle \Pi \rangle \mathcal{O}_{\mathbb{L}}^2$, where $U^{-1} J_2 t_* U$ is computed from $\mathbf{Q}$ via Proposition 2.1. Assuming $u_1 = U v_1$, we have $u_1 \not\equiv (0, 0)^T, (1, 1)^T \pmod{\langle \Pi \rangle \mathcal{O}_{\mathbb{L}}^2}$; otherwise, $U^{-1}(J_2 t_* u_0 - u_0) \equiv (0, 0) \pmod{\langle \Pi \rangle \mathcal{O}_{\mathbb{L}}^2}$. Without loss of generality, we can assume $u_1 = (\alpha_1, \beta_1)$ with $\alpha_1 \in 1 + \langle \Pi \rangle$ and $\beta_1 \in \langle \Pi \rangle$; otherwise, we replace U with $J_2 U$.

Suppose we have computed v_i such that $u_i := U v_i = (\alpha_i, \beta_i)$, where $\alpha_i \in 1 + \langle \Pi \rangle$ and $\beta_i \in \langle \Pi \rangle^i$ for some $i \geq 1$. Assume further that $\beta_i / \Pi^i \equiv c \pmod{\langle \Pi \rangle}$ for some $c \in \{0, 1\}$. Consider the free module lattice M defined by

$$\begin{aligned} M &= v_i \cdot \mathcal{O}_{\mathbb{L}} + \langle \Pi^{i+1} \rangle \mathcal{O}_{\mathbb{L}}^2 \\ &= U^{-1}(u_i \cdot \mathcal{O}_{\mathbb{L}} + \langle \Pi^{i+1} \rangle \mathcal{O}_{\mathbb{L}}^2) \\ &= U^{-1}((1, c\Pi^i)^T \cdot \mathcal{O}_{\mathbb{L}} + (0, \Pi^{i+1})^T \cdot \mathcal{O}_{\mathbb{L}}) \\ &= U^{-1}B_{i+1}^{(c)} \cdot \mathcal{O}_{\mathbb{L}}^2. \end{aligned}$$

Regarding the justification for the third equality. The left-hand side is contained in the right-hand side: since $c \cdot \alpha_i \Pi^i - \beta_i \in \langle \Pi \rangle^{i+1}$, there exists $t_i \in \mathcal{O}_{\mathbb{L}}$ such that

$$(1, c\Pi^i)^T \cdot \alpha_i - (0, \Pi^{i+1})^T \cdot t_i = u_i,$$

and

$$(H^{i+1}, 0)^T = (1, cH^i)^T \cdot H^{i+1} - (0, H^{i+1})^T \cdot cH^i.$$

The right-hand side is contained in the left-hand side: it suffices to show that $(1, cH^i)^T$ belongs to the left-hand side. Note that there exists $w_i \in 1 + \langle H \rangle$ such that $w_i \alpha_i - 1 \in \langle H^{i+1} \rangle$ (for example, take $w_i = \sum_{j=0}^i (1 - \alpha_i)^j$). Moreover, there exists $s_i \in \mathcal{O}_{\mathbb{L}}$ such that $w_i \beta_i - cH^i = s_i H^{i+1}$. Hence,

$$(1, cH^i)^T = u_i \cdot w_i - (0, H^{i+1})^T \cdot s_i.$$

Using v_i , we can compute a $\mathbb{Z}$ -basis of M . Then, by [Lemma 2.6](#), we can compute a free basis $B = U^{-1}B_{i+1}^{(c)}V$ for some $V \in \text{GL}_2(\mathcal{O}_{\mathbb{L}})$ in polynomial time on a quantum computer. Taking $\mathbf{Q}_{i+1} = B^* \mathbf{Q} B$ as the input to $\Delta_{i+1, \mathbb{L}}$, we can obtain the value of c .

Let $\tilde{v}_i = U^{-1}J_2 t_* U v_i$. Define $v_{i+1} = v_i + cH^i \cdot \tilde{v}_i$. Assuming $U v_{i+1} = (\alpha_{i+1}, \beta_{i+1})$, note that

$$U v_{i+1} = U v_i + cH^i J_2 t_* U v_i = (\alpha_i + cH^i \beta_i^*, \beta_i - cH^i \alpha_i^*).$$

Using the facts that $\alpha_i \in 1 + \langle H \rangle$ and $\beta_i/H^i \equiv c \pmod{\langle H \rangle}$, we obtain $\alpha_{i+1} \in 1 + \langle H \rangle$ and $\beta_{i+1} \in \langle H \rangle^{i+1}$.

Inductively, we obtain v_{2n^2} . Consider the module lattice $M = v_{2n^2} \cdot \mathcal{O}_{\mathbb{L}} + \langle H^{2n^2} \rangle \mathcal{O}_{\mathbb{L}}^2$. Similar to the previous argument, we have $M = U^{-1}B_{2n^2}^{(0)} \mathcal{O}_{\mathbb{L}}^2 = U^{-1}((1, 0)^T \cdot \mathcal{O}_{\mathbb{L}} \oplus (0, 2^{2n})^T \cdot \mathcal{O}_{\mathbb{L}})$. Compute an LLL-reduced basis (b_i) of M under the norm induced by $\mathbf{Q}$; then b_1 satisfies $\|U b_1\| \leq 2^{(2n-1)/2} \lambda_1(U \cdot M) = \sqrt{n} 2^{(2n-1)/2}$. Assume that $U b_1 = (a, 2^{2n} b)^T$ for some $a, b \in \mathcal{O}_{\mathbb{L}}$. Then $\|U b_1\|^2 = \text{tr}(a^* a) + \text{tr}(2^{4n} b^* b)$, and from the previous upper bound we deduce $b = 0$. We compute

$$M_1 = b_1 \cdot \mathbb{L} \cap \mathcal{O}_{\mathbb{L}}^2 = U^{-1}((a, 0)^T \cdot \mathbb{L} \cap U \cdot \mathcal{O}_{\mathbb{L}}^2) = (U^{-1}(1, 0)^T) \cdot \mathcal{O}_{\mathbb{L}}$$

(by computing $\text{span}_{\mathbb{Q}}(\text{rot}(b_1)) \cap \mathbb{Z}^{2n}$). Using [Proposition 2.2](#), input $\mathcal{O}_{\mathbb{L}}$, $\mathbf{Q}$, 1, and M_1 into LS2 to obtain $U^{-1}(1, 0)^T \cdot \mu(\mathbb{L})$. Apply $U^{-1}J_2 t_* U$ to get $U^{-1}(0, 1)^T \cdot \mu(\mathbb{L})$. By enumerating $\mu(\mathbb{L})$ (which can be done in polynomial time according to [Lemma 2.4](#)), we can compute U^{-1} . $\square$

Lemma C.1 $\forall x \in \mathfrak{p}$, $\|x\|^2 \in 2n\mathbb{Z}$. $\forall x \in \mathcal{O}_{\mathbb{L}}$, $\|x\|^2 = 2n$ iff $x = (1 + X^s)X^t$ for some $s, t \in \mathbb{Z}$. If $x_1 = (1 + X^{s_1})X^{t_1}$ and $x_2 = (1 + X^{s_2})X^{t_2}$ s.t. $x_1^* x_1 = x_2^* x_2$, then $x_1/x_2 \in \mu(\mathbb{L})$.

Proof. Note that

$$\frac{1}{n} \|x\|^2 = \|\text{vec}(x)\|^2 \equiv \left\langle \text{vec}(x), \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \right\rangle \pmod{2} = \frac{1}{n} \langle x, 1 + X + \cdots + X^{n-1} \rangle.$$

Since $x \in \mathfrak{p}$, we can write x as $b(1 - X^{-1})$ for some $b \in \mathcal{O}_{\mathbb{L}}$. Then

$$\begin{aligned} \frac{1}{n} \langle x, 1 + X + \cdots + X^{n-1} \rangle &= \frac{1}{n} \text{Tr}_{\mathbb{L}}(b^*(1 - X)(1 + X + \cdots + X^{n-1})) \\ &= 2 \frac{1}{n} \text{Tr}_{\mathbb{L}}(b^*) \in 2\mathbb{Z}. \end{aligned}$$

Thus we have $\|x\|^2 \in 2n\mathbb{Z}$ and $\|x\|^2 = 2n$ iff $\|\text{vec}(x)\|^2 = 2$ iff $\text{vec}(x)$ has only two non-zero components, and they belong to ± 1 iff $x = (1 + X^s)X^t$ for some $s, t \in \mathbb{Z}$. If $x_1 = (1 + X^{s_1})X^{t_1}$ and $x_2 = (1 + X^{s_2})X^{t_2}$ s.t. $x_1^*x_1 = x_2^*x_2$, we have

$$(1 + X^{s_1})^*(1 + X^{s_1}) = (1 + X^{s_2})^*(1 + X^{s_2}),$$

which implies that $X^{s_1} + X^{-s_1} = X^{s_2} + X^{-s_2}$. Therefore, we can deduce that $\{X^{s_1}, X^{-s_1}\} = \{X^{s_2}, X^{-s_2}\}$ since they are both the root sets of the equation $x^2 - (X^{s_1} + X^{-s_1})x + 1 = 0$. It follows that $1 + X^{s_1} = 1 + X^{s_2}$ or $1 + X^{-s_2} = X^{-s_2}(1 + X^{s_2})$, and thus $x_1/x_2 \in \mu(\mathbb{L})$. $\square$

Proof (of Lemma 3.1). When $U = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GSL}_2(\mathcal{O}_{\mathbb{L}})$ such that $(\mathbf{B}_k^{(0)}U)^*(\mathbf{B}_k^{(0)}U) = (\mathbf{B}_k^{(0)})^*\mathbf{B}_k^{(0)}$, we first have

$$\left\| \begin{pmatrix} a \\ c(1 + X)^k \end{pmatrix} \right\|^2 = \left\| \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\|^2 = n.$$

By Lemma C.1, $\|c(1 + X)^k\|^2 \in 2n\mathbb{Z}$. Note that $\|a\|^2 = n \langle \text{vec}(a), \text{vec}(a) \rangle \in n\mathbb{Z}$. Thus $c = 0$ and $a \in \mu(\mathbb{L})$. We also have

$$b^*a = \begin{pmatrix} b \\ d(1 + X)^k \end{pmatrix}^* \begin{pmatrix} a \\ c(1 + X)^k \end{pmatrix} = \begin{pmatrix} 0 \\ (1 + X)^k \end{pmatrix}^* \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0.$$

Obviously, $ad = ad - bc \in \mu(\mathbb{L})$ implies that $d \in \mu(\mathbb{L})$. Thus

$$\text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q}_k^{(0)}) \subseteq \left\{ \begin{pmatrix} \xi_1 & 0 \\ 0 & \xi_2 \end{pmatrix} \mid \xi_1, \xi_2 \in \mu(\mathbb{L}) \right\}.$$

The reverse inclusion follows by direct verification. $\square$

Proof (of Lemma 3.2). Note

$$\mathbf{B}_k^{(2)} = \begin{pmatrix} 1 & 0 \\ (1 + X^2)^{k-1} & (1 + X^2)^k \end{pmatrix} = \begin{pmatrix} 1 & 1 + X^2 \\ (1 + X^2)^{k-1} & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 + X^2 \\ 0 & -1 \end{pmatrix}.$$

Denote $\begin{pmatrix} 1 & 1 + X^2 \\ (1 + X^2)^{k-1} & 0 \end{pmatrix}$ by $\tilde{\mathbf{B}}_k^{(2)}$. It suffices to show that

$$\text{Aut}_{\mathcal{O}_{\mathbb{L}}}((\tilde{\mathbf{B}}_k^{(2)})^*\tilde{\mathbf{B}}_k^{(2)}) = \left\{ \begin{pmatrix} \xi_1 & 0 \\ \frac{\xi_2 - \xi_1}{1 + X^2} & \xi_2 \end{pmatrix} \mid \xi_1, \xi_2 \in \mu(\mathbb{L}), \xi_1/\xi_2 = X^{2j}, j \in \mathbb{Z} \right\}.$$

If there exists a matrix $U = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GSL_2(\mathcal{O}_{\mathbb{L}})$ such that

$$(\widetilde{\mathbf{B}}_k^{(2)}U)^*(\widetilde{\mathbf{B}}_k^{(2)}U) = (\widetilde{\mathbf{B}}_k^{(2)})^*\widetilde{\mathbf{B}}_k^{(2)},$$

we first have

$$\left\| \begin{pmatrix} b + d(1 + X^2) \\ b(1 + X^2)^{k-1} \end{pmatrix} \right\|^2 = \left\| \begin{pmatrix} 1 + X^2 \\ 0 \end{pmatrix} \right\|^2 = 2n.$$

By [Lemma C.1](#), $\|b(1 + X^2)^{k-1}\|^2 \in 2n\mathbb{Z}$. Thus we have

$$\|b(1 + X^2)^{k-1}\|^2 = 0, \|b + d(1 + X^2)\| = 2n$$

or

$$\|b(1 + X^2)^{k-1}\|^2 = 2n, \|b + d(1 + X^2)\| = 0.$$

Note that we further have

$$\begin{pmatrix} b + d(1 + X^2) \\ b(1 + X^2)^{k-1} \end{pmatrix}^* \begin{pmatrix} b + d(1 + X^2) \\ b(1 + X^2)^{k-1} \end{pmatrix} = \begin{pmatrix} 1 + X^2 \\ 0 \end{pmatrix}^* \begin{pmatrix} 1 + X^2 \\ 0 \end{pmatrix} = (1 + X^2)^*(1 + X^2),$$

and thus by [Lemma C.1](#), we have either

$$b(1 + X^2)^{k-1} = 0, \quad b + d(1 + X^2) = (1 + X^2)\xi_2,$$

or

$$b(1 + X^2)^{k-1} = (1 + X^2)\xi_2, \quad b + d(1 + X^2) = 0$$

for some $\xi_2 \in \mu(\mathbb{L})$.

For the second case, since $k > 2$ and $1 + X^2$ is not a unit in $\mathcal{O}_{\mathbb{L}}$, a contradiction arises. So we have $b = 0$, $d = \xi_2$. Moreover,

$$\begin{aligned} \xi_2^*(1 + X^2)^*(a + c(1 + X^2)) &= \begin{pmatrix} b + d(1 + X^2) \\ b(1 + X^2)^k \end{pmatrix}^* \begin{pmatrix} a + c(1 + X^2) \\ a(1 + X^2)^{k-1} \end{pmatrix} \\ &= \begin{pmatrix} 1 + X^2 \\ 0 \end{pmatrix}^* \begin{pmatrix} 1 \\ (1 + X^2)^{k-1} \end{pmatrix} \\ &= (1 + X^2)^*. \end{aligned}$$

Thus, $a + c(1 + X^2) = \xi_2$. Since $ad - bc \in \mu(\mathbb{L})$ and $b = 0$, it follows that $ad \in \mu(\mathbb{L})$, which implies $a \in \mu(\mathbb{L})$. Denote a as ξ_1 . Then we have $c = \frac{\xi_2 - \xi_1}{1 + X^2}$. Since $c \in \mathcal{O}_{\mathbb{L}}$, we have $1 - \xi_1/\xi_2 \in \langle 1 + X^2 \rangle = \mathfrak{p}^2 \Rightarrow \xi_1/\xi_2 = X^{2k}$ for some $k \in \mathbb{Z}$ (otherwise $\langle 1 - \xi_1/\xi_2 \rangle = \mathfrak{p}$, there is a contradiction). Therefore

$$\text{Aut}_{\mathcal{O}_{\mathbb{L}}}((\widetilde{\mathbf{B}}_k^{(2)})^*\widetilde{\mathbf{B}}_k^{(2)}) \subseteq \left\{ \begin{pmatrix} \xi_1 & 0 \\ \frac{\xi_2 - \xi_1}{1 + X^2} & \xi_2 \end{pmatrix} \mid \xi_1, \xi_2 \in \mu(\mathbb{L}), \xi_1/\xi_2 = X^{2k}, k \in \mathbb{Z} \right\}.$$

The reverse inclusion follows by direct verification. $\square$

Proof (of Theorem 3.3). We will complete the proof in the following order. First, we prove that for all $\text{poly}(n) \geq \ell \geq k$,

$$\text{module-LIP}_{[\mathbf{Q}_k^{(0)}]} \geq \text{module-LIP}_{[\mathbf{Q}_\ell^{(0)}]} \geq \text{module-}\Delta\text{LIP}_{\mathcal{D}_{[\mathbf{Q}_\ell^{(0)}]}, \mathcal{D}_{[\mathbf{Q}_\ell^{(1)}]}}.$$

Then, for the decision oracles required to solve $\mathcal{O}_{\mathbb{L}}^2$ -LIP in Theorem 2.1, we implement the solver for $\text{module-}\Delta\text{LIP}_{\mathcal{D}_{[\mathbf{Q}_\ell^{(0)}]}, \mathcal{D}_{[\mathbf{Q}_\ell^{(1)}]}}$ with $\ell \geq k$ using the oracle for $\text{module-LIP}_{[\mathbf{Q}_k^{(0)}]}$, while for $\ell < k$ we simply guess.

Since $k = O(\log n)$, as long as we have access to the oracle for $\text{module-LIP}_{[\mathbf{Q}_k^{(0)}]}$, we can provide all oracle outputs required by Theorem 2.1 to solve $\mathcal{O}_{\mathbb{L}}^2$ -LIP in polynomial time with probability $1/2^k$, thereby achieving a successful solution.

The reduction $\text{module-LIP}_{[\mathbf{Q}_\ell^{(0)}]} \geq \text{module-}\Delta\text{LIP}_{\mathcal{D}_{[\mathbf{Q}_\ell^{(0)}]}, \mathcal{D}_{[\mathbf{Q}_\ell^{(1)}]}}$ is straightforward. One simply needs to feed the instance of $\text{module-}\Delta\text{LIP}_{\mathcal{D}_{[\mathbf{Q}_\ell^{(0)}]}, \mathcal{D}_{[\mathbf{Q}_\ell^{(1)}]}}$ to the solver for $\text{module-LIP}_{[\mathbf{Q}_\ell^{(0)}]}$. If the solver succeeds, output 0; otherwise, output 1. Thus, it remains only to prove that for any $\text{poly}(n) \geq \ell \geq k$, $\text{module-LIP}_{[\mathbf{Q}_k^{(0)}]} \geq \text{module-LIP}_{[\mathbf{Q}_\ell^{(0)}]}$.

Given $U_0^* \mathbf{Q}_0^{(\ell)} U_0$, we first iteratively compute matrices M_i such that there exists $U_{i+1} \in \text{GL}_2(\mathcal{O}_{\mathbb{L}})$ satisfying

$$M_i = U_i^{-1} \begin{pmatrix} \Pi & 0 \\ 0 & 1 \end{pmatrix} U_{i+1}, \quad i = 0, \dots, \ell - k - 1.$$

Define $\mathbf{G}_0 = U_0^* \mathbf{Q}_0^{(0)} U_0$ and $\mathbf{G}_i = \Pi^{-2} M_i^* \mathbf{G}_{i-1} M_i$. Then we have $\mathbf{G}_i = U_i^* \mathbf{Q}_{\ell-i}^{(0)} U_i$. Suppose M_{i-1} has been computed (let it be nothing if $i = 0$) and $\mathbf{G}_i$ is also known. Consider the set

$$\begin{aligned} \mathcal{L}_i &= \{x \in \mathcal{O}_{\mathbb{L}}^2 \mid x^* \mathbf{G}_i x \in \mathfrak{p}\} = U_i^{-1} \{x \in \mathcal{O}_{\mathbb{L}}^2 \mid x^* \mathbf{Q}_{\ell-i}^{(0)} x \in \mathfrak{p}\} \\ &= U_i^{-1} \left(\mathcal{O}_{\mathbb{L}} \begin{pmatrix} \Pi \\ 0 \end{pmatrix} + \mathcal{O}_{\mathbb{L}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \end{aligned}$$

is a free $\mathcal{O}_{\mathbb{L}}$ module. We first compute a $\mathbb{Z}$ -basis of $\mathcal{L}_i$, and then compute the $\mathcal{O}_{\mathbb{L}}$ -basis M_i of $\mathcal{L}_i$ by Lemma 2.6 in polynomial time on a quantum computer. By the property of free bases, there exists $U_{i+1} \in \text{GL}_2(\mathcal{O}_{\mathbb{L}})$ such that

$$M_i = U_i^{-1} \begin{pmatrix} \Pi & 0 \\ 0 & 1 \end{pmatrix} U_{i+1}.$$

After computing $M_i = U_i^{-1} \begin{pmatrix} \Pi & 0 \\ 0 & 1 \end{pmatrix} U_{i+1}$ for $i = 0, \dots, \ell - k - 1$ and obtaining $\mathbf{G}_{\ell-k}$, inputting $\mathbf{G}_{\ell-k}$ into the solver for $\text{module-LIP}_{[\mathbf{Q}_k^{(0)}]}$ will output an element $T_{\ell-k}$ of $U_{\ell-k}^{-1} \cdot \text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q}_k^{(0)})$. For $k > 0$, $T_{\ell-k} = U_{\ell-k}^{-1} \begin{pmatrix} \xi_1 & 0 \\ 0 & \xi_2 \end{pmatrix}$ for some $\xi_1, \xi_2 \in \mu(\mathbb{L})$.

For $k = 0$, we consider both the case of $T_{\ell-k}$ and the case of replacing $T_{\ell-k}$ with $T_{\ell-k}J_2$. In summary, we may assume that $T_{\ell-k}$ is of the form $U_{\ell-k}^{-1} \begin{pmatrix} \xi_1 & 0 \\ 0 & \xi_2 \end{pmatrix}$.

By recursively computing $T_i = M_i T_{i+1} \begin{pmatrix} \Pi & 0 \\ 0 & 1 \end{pmatrix}^{-1}$, we can recursively prove that $T_i = U_i^{-1} \begin{pmatrix} \xi_1 & 0 \\ 0 & \xi_2 \end{pmatrix}$ (utilizing the commutativity of diagonal matrix multiplication) for $i = \ell - k, \dots, 0$. Then T_0 is an isomorphism from $\mathbf{G}_0$ to $\mathbf{Q}_\ell^{(0)}$.

Regarding the computation of $\mathcal{L}_i$, there are multiple approaches. One can utilize the identity

$$(x + y)^* \mathbf{G}_i(x + y) = x^* \mathbf{G}_i x + y^* \mathbf{G}_i y \pmod{\mathfrak{p}}$$

(since for any $z \in \mathcal{O}_\mathbb{L}$, $z + z^* \in \mathfrak{p}$) to transform it into solving a linear equation over $\mathbb{Z}_2$. Alternatively, one can repeatedly perform Gaussian sampling until vectors falling in $\mathcal{L}_i$ can generate $\mathcal{L}_i$ (determined by checking the determinant). $\square$

Proof (of Lemma 4.1). By Lemma 2.1, the first step can be performed efficiently. By Lemma 2.2, Algorithm `HermiteSolve` is efficient. By Heuristic 1, the expected number of iterations for Steps 2–6 is approximately $4/3$. Therefore, this is an efficient randomized algorithm.

Next, we consider the distribution of $R_0(\mathbf{Q})[2]$. For any $\mathbf{U} = (u_1|u_2) \in \text{SL}_2(\mathcal{O}_\mathbb{L})$, let $\widetilde{u}_2 = \text{HermiteSolve}(\mathbb{L}, u_1)$ and $u'_2 = \text{Reduce}(\mathbf{Q}, u_1, \widetilde{u}_2)$. Then, from the steps of the algorithm, it follows that the probability of sampling $\mathbf{U}$ equals the probability of obtaining u_1 via rejection sampling in Steps 1–4, multiplied by the probability that u_2 equals the value u'_2 computed from the inputs f, g in Steps 5–7, that is,

$$\Pr[R_0(\mathbf{Q})[2] = \mathbf{U}] = D_{\mathbf{Q}, \sigma}(u_1) / D_{\mathbf{Q}, \sigma}((\mathcal{O}_\mathbb{L}^2)^*) \cdot \Pr[u_2 = u'_2].$$

The first item is equal to $\rho_{\mathbf{Q}, \sigma}(u_1) / \rho_{\mathbf{Q}, \sigma}((\mathcal{O}_\mathbb{L}^2)^*) = \rho_{\mathbf{Q}, \sigma}(u_1) / \rho_\sigma((\mathcal{O}_\mathbb{L}^2)^*)$. It remains to show $u_2 = u'_2$ iff u_2 is reduced with respect to u_1 and $\mathbf{Q}$.

If $u_2 = u'_2 = \text{Reduce}(\mathbf{Q}, u_1, \widetilde{u}_2) = \widetilde{u}_2 - \left\lceil \frac{u_1^* \cdot \mathbf{Q} \cdot \widetilde{u}_2}{u_1^* \cdot \mathbf{Q} \cdot u_1} \right\rceil \cdot u_1$, then

$$\left\lceil \frac{u_1^* \cdot \mathbf{Q} \cdot u_2}{u_1^* \cdot \mathbf{Q} \cdot u_1} \right\rceil = \left\lceil \frac{u_1^* \cdot \mathbf{Q} \cdot \widetilde{u}_2}{u_1^* \cdot \mathbf{Q} \cdot u_1} - \left\lceil \frac{u_1^* \cdot \mathbf{Q} \cdot \widetilde{u}_2}{u_1^* \cdot \mathbf{Q} \cdot u_1} \right\rceil \right\rceil = 0.$$

Conversely, assume that $\left\lceil \frac{u_1^* \cdot \mathbf{Q} \cdot u_2}{u_1^* \cdot \mathbf{Q} \cdot u_1} \right\rceil = 0$. By previous argument, $\left\lceil \frac{u_1^* \cdot \mathbf{Q} \cdot u'_2}{u_1^* \cdot \mathbf{Q} \cdot u_1} \right\rceil$ is also 0. Note $(u_1|u_2), (u_1|u'_2)$ are both in $\text{SL}_2(\mathcal{O}_\mathbb{L})$, so $\det((u_1|u_2 - u'_2)) = 0$. Hence there exists some $\lambda \in \mathbb{L}$ s.t. $u_2 - u'_2 = \lambda u_1$. Since $u_1 \in (\mathcal{O}_\mathbb{L}^2)^*$ and $u_2, u'_2 \in \mathcal{O}_\mathbb{L}^2$, we have

$$(\lambda) = (\lambda u_1[1]) + (\lambda u_1[2]) = ((u_2 - u'_2)[1]) + ((u_2 - u'_2)[2]) \subseteq \mathcal{O}_\mathbb{L}.$$

This means $\lambda \in \mathcal{O}_\mathbb{L}$. Thus $0 = \left\lceil \frac{u_1^* \cdot \mathbf{Q} \cdot (u_2 - u'_2)}{u_1^* \cdot \mathbf{Q} \cdot u_1} \right\rceil = \lceil \lambda \rceil = \lambda$, i.e. $u_2 = u'_2$.

Finally, for any $\mathbf{U} = (u_1|u_2) \in \text{SL}_2(\mathcal{O}_{\mathbb{L}})$, we have

$$\Pr[\mathbf{V}^{-1} \cdot \mathbf{R}_0(\mathbf{Q})[2] = \mathbf{U}] = \Pr[\mathbf{R}_0(\mathbf{Q})[2] = \mathbf{V}\mathbf{U}].$$

Using the distribution expression derived above, we obtain

$$\Pr[\mathbf{R}_0(\mathbf{Q})[2] = \mathbf{V}\mathbf{U}] = \begin{cases} \rho_{\mathbf{Q},\sigma}(\mathbf{V}u_1)/\rho_{\mathbf{Q}_0,\sigma}((\mathcal{O}_{\mathbb{L}}^2)^*), & \text{if } \mathbf{V}u_2 \text{ is reduced with respect} \\ & \text{to } \mathbf{V}u_1 \text{ and } \mathbf{Q}; \\ 0, & \text{otherwise,} \end{cases}$$

and

$$\Pr[\mathbf{R}_0(\mathbf{V}^*\mathbf{Q}\mathbf{V})[2] = \mathbf{U}] = \begin{cases} \rho_{\mathbf{V}^*\mathbf{Q}\mathbf{V},\sigma}(u_1)/\rho_{\mathbf{Q}_0,\sigma}((\mathcal{O}_{\mathbb{L}}^2)^*), & \text{if } u_2 \text{ is reduced with respect} \\ & \text{to } u_1 \text{ and } \mathbf{V}^*\mathbf{Q}\mathbf{V}; \\ 0, & \text{otherwise.} \end{cases}$$

Note $\rho_{\mathbf{Q},\sigma}(\mathbf{V}u_1) = \rho_{\mathbf{V}^*\mathbf{Q}\mathbf{V},\sigma}(u_1)$. We just need to show $\mathbf{V}u_2$ is reduced with respect to $\mathbf{V}u_1$ and $\mathbf{Q}$ iff u_2 is reduced with respect to u_1 and $\mathbf{V}^*\mathbf{Q}\mathbf{V}$. This follows directly from the definition. $\square$

Appendix D Discussion about extractors over $\text{SL}_2(\mathcal{O}_{\mathbb{L}})$

In this section, we focus on proving the following theorem. Regarding the extractor as a group homomorphism is motivated by two factors: first, it is required for constructing an enhanced linkable commitment; second, it represents a natural methodology for extractor construction itself. Our proof reveals that the feasibility of finding such an extractor is intimately connected to the structure of the stabilizer subgroup of the group action—specifically, the automorphism group of a module lattice. In this section, we consider the number field $\mathbb{L} = \mathbb{Q}[X]/(X^{2^{d-1}} + 1)$, where $d > 3$.

Theorem D.1 *For the group action $(\text{SL}_2(\mathcal{O}_{\mathbb{L}}), [\mathbf{Q}_{\mathbf{k}}^{(0)}]_{sl}, \star)$, $k \geq 0$, there exists no extractor satisfying the **uniformly extracting property and homomorphic property**.*

To complete the proof, we need to introduce some structural properties of $\text{SL}_2(\mathcal{O}_{\mathbb{L}})$ and $\text{Aut}_{\mathcal{O}_{\mathbb{L}}}(\mathbf{Q}_{\mathbf{k}}^{(0)})$.

Definition D.1 *For R a unital ring, we denote by $E_n(R)$ the subgroup of $GL_n(R)$ generated by all transvections $e_{ij}(r) = I_n + r\epsilon_{ij}$ (a.k.a. the elementary matrices over R) with $r \in R$, $1 \leq i \neq j \leq n$, where I_n is the identity matrix and ϵ_{ij} is the matrix whose (i, j) entry is 1 while all its other entries are zero.*

Lemma D.1 (induced by [32]) *$\text{SL}_2(\mathcal{O}_{\mathbb{L}}) = E_2(\mathcal{O}_{\mathbb{L}})$. Here we make use of the fact that $\mathcal{O}_{\mathbb{L}}^\times$ is infinite.*

From this, we easily obtain the following lemma.

Lemma D.2 *The derived subgroup of $SL_2(\mathcal{O}_{\mathbb{L}})$ is itself, i.e. $[SL_2(\mathcal{O}_{\mathbb{L}}), SL_2(\mathcal{O}_{\mathbb{L}})] = SL_2(\mathcal{O}_{\mathbb{L}})$.*

Proof. Since $e_{ik}(s) = [e_{ij}(1), e_{jk}(s)] \in [SL_2(\mathcal{O}_{\mathbb{L}}), SL_2(\mathcal{O}_{\mathbb{L}})]$, $\forall s \in \mathcal{O}_{\mathbb{L}}$, $E_2(\mathcal{O}_{\mathbb{L}}) \subseteq [SL_2(\mathcal{O}_{\mathbb{L}}), SL_2(\mathcal{O}_{\mathbb{L}})]$. By Lemma D.1, $SL_2(\mathcal{O}_{\mathbb{L}}) = E_2(\mathcal{O}_{\mathbb{L}}) \subseteq [SL_2(\mathcal{O}_{\mathbb{L}}), SL_2(\mathcal{O}_{\mathbb{L}})]$. So $[SL_2(\mathcal{O}_{\mathbb{L}}), SL_2(\mathcal{O}_{\mathbb{L}})] = SL_2(\mathcal{O}_{\mathbb{L}})$. $\square$

Lemma D.3 *Let $g = \begin{pmatrix} \xi & 0 \\ 0 & \xi^{-1} \end{pmatrix}$. For*

$$SU_2(\mathcal{O}_{\mathbb{L}}) := \left\{ \begin{pmatrix} \xi & 0 \\ 0 & \xi^{-1} \end{pmatrix} \mid \xi \in \mu(\mathbb{L}) \right\} \cup \left\{ \begin{pmatrix} 0 & \xi \\ -\xi & 0 \end{pmatrix} \mid \xi \in \mu(\mathbb{L}) \right\} = \langle g, J_2 \rangle$$

, the normal subgroups of $SU_2(\mathcal{O}_{\mathbb{L}})$ include $\langle g^{2^i} \rangle$ for $i \in \mathbb{Z}$, as well as $\langle J_2, g^2 \rangle$, $\langle gJ_2, g^2 \rangle$ and $SU_2(\mathcal{O}_{\mathbb{L}})$ itself. Moreover, the derived subgroup $[SU_2(\mathcal{O}_{\mathbb{L}}), SU_2(\mathcal{O}_{\mathbb{L}})]$ of $SU_2(\mathcal{O}_{\mathbb{L}})$ is $\langle g^2 \rangle$.

Proof. The key observation is that $gJ_2 = J_2g^{-1}$. Let H be a normal subgroup of $SU_2(\mathcal{O}_{\mathbb{L}})$. If H contains no anti-diagonal matrix, then $H \subseteq \langle g \rangle$ is isomorphic to $C_{2^{d-1}}$, which is also cyclic and generated by some $\begin{pmatrix} X^{2^i} & 0 \\ 0 & X^{-2^i} \end{pmatrix}$. If H contains an anti-diagonal matrix, which can be written as $g^k J_2$, then $g^{k+2} J_2 = gg^k J_2 g^{-1} \in H$, and consequently $g^2 = g^{k+2} J_2 (g^k J_2)^{-1} \in H$. If H contains both J_2 and g , then $H = SU_2(\mathcal{O}_{\mathbb{L}})$; if H contains J_2 but not g , then $H = \langle g^2, J_2 \rangle$; if H contains neither J_2 nor g , then k is odd, and consequently $H = \langle gJ_2, g^2 \rangle$. Among these normal subgroups, $\langle g^2 \rangle$ is the minimal normal subgroup that makes the quotient group abelian; therefore, $[SL_2(\mathcal{O}_{\mathbb{L}}), SL_2(\mathcal{O}_{\mathbb{L}})] = SL_2(\mathcal{O}_{\mathbb{L}})$. $\square$

Proof (of Theorem D.1). Suppose such an extractor $E : SL_2(\mathcal{O}_{\mathbb{L}}) \rightarrow M$ exists. It is a group homomorphism, and the uniformly extracting property is equivalent to the restriction of E to $\text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)})$ being surjective. Continuing with $g = \begin{pmatrix} X & 0 \\ 0 & X^{-1} \end{pmatrix}$, by Lemma 3.1, for $k > 0$, $\text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)}) = \langle g \rangle$; and by Lemma 2.3, $\text{Stab}(\mathbf{Q}_0^{(0)}) = SU_2(\mathcal{O}_{\mathbb{L}})$.

We have the following equality:

$$\begin{aligned} M &= E(SL_2(\mathcal{O}_{\mathbb{L}})) = E([SL_2(\mathcal{O}_{\mathbb{L}}), SL_2(\mathcal{O}_{\mathbb{L}})]) = [E(SL_2(\mathcal{O}_{\mathbb{L}})), E(SL_2(\mathcal{O}_{\mathbb{L}}))] \\ &= [E(\text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)})), E(\text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)}))] = E([\text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)}), \text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)})]). \end{aligned}$$

For $k > 0$, the $[\text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)}), \text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)})]$ is trivial, while $|M| > 1$, a contradiction. For $k = 0$, by the previous lemma, note that any normal subgroup of $\text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)})$ either contains $[\text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)}), \text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)})]$ or is contained in $[\text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)}), \text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)})]$. Therefore, either $\ker(E|_{\text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)})}) = \text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)})$ or $E([\text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)}), \text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)})])$ is strictly contained in $E(SL_2(\mathcal{O}_{\mathbb{L}}))$. The former implies that M is trivial, a contradiction; the latter contradicts the fact that $M = E([\text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)}), \text{Stab}(\mathbf{Q}_{\mathbf{k}}^{(0)})])$. $\square$